regc — Replica-Exchange Grand-Canonical Monte Carlo

User manual: physics, design, workflow and worked examples

November 2026

Contents

1. The physics . . . 1
2. How the package is organised . . . 8
3. A calculation step by step . . . 9
4. Worked examples . . . 11
5. Analysis in depth . . . 17
6. Running from a config file . . . 25
7. Long runs on a cluster . . . 26
8. Good practice and pitfalls . . . 27
9. References . . . 28

Appendix A. Figure provenance . . . 29

`regc` computes **surface and nanoparticle phase diagrams at finite temperature and gas pressure**. It samples a two-dimensional ladder of grand-canonical replicas over temperature T and chemical potential μ, exchanging configurations between them, and reweights all samples with MBAR into smooth maps of coverage, heat capacity and structure as functions of (T, p).

The method is replica-exchange grand-canonical Monte Carlo (REGC) of Zhou, Scheffler and Ghiringhelli [ZSG19, ZSG22]. It is anharmonic: no harmonic vibrational free energy is assumed, and configurational entropy is sampled rather than enumerated.

This manual explains the physics behind each step, how the package is organised, and how to run, analyse and trust a calculation. The worked examples and figures were produced with the package itself; the script `docs/manual/make_figures.py` regenerates every figure.

The reference manual lists every option. The algorithm notes give every acceptance rule with its derivation. The API reference documents every public function.

---

1. The physics

1.1 The question

A surface in contact with a gas at temperature T and pressure p takes up gas atoms until the chemical potential of the adsorbed atoms equals that of the gas. Which structure is stable is decided by a free energy. That free energy includes energy, configurational entropy (where the adsorbates sit) and vibrational entropy (how they and the substrate move). Ab initio atomistic thermodynamics (aiAT) compares a few hand-picked structures, with harmonic vibrations at best. REGC instead samples the grand-canonical ensemble directly, so every structure the potential allows competes, with all anharmonic effects.

1.2 The grand-canonical ensemble

A replica at (T, μ) samples configurations **R** of N exchangeable atoms with the probability density [ZSG19, Eq. 1]

$$\mathscr{P}(\mathbf{R}, N) \;\propto\; \frac{V^N}{N!\,\Lambda^{3N}(T)}\, e^{\beta\mu N}\, e^{-\beta U(\mathbf{R},N)}, \qquad \beta = \frac{1}{k_B T}, \qquad \Lambda = \frac{h}{\sqrt{2\pi m k_B T}}. \tag{1}$$

- **U** is the potential energy from any calculator: an EAM, an MLIP, or LJ.
- **The substrate** (the slab or cluster atoms) is part of every configuration, partly frozen (the bottom layers of a slab, the core of a cluster) and partly free to move.
- **The kinetic energy** has been integrated out. It is restored analytically where it matters, in the heat capacity (§1.9).

All energies are on the calculator's own energy scale. An isolated gas atom therefore has energy `e_atom`, not 0, and μ must be on the same scale (§1.3).

1.3 The gas reservoir: from (T, p) to μ

Experiments control T and p; the simulation needs μ of the exchanged **atom**. For a monatomic ideal gas,

$$\mu(T, p) = k_B T\, \ln \frac{p\,\Lambda^3}{k_B T} + e_{\text{atom}}. \tag{2}$$

Most reactive gases are diatomic (H_2, O_2, N_2, Cl_2), yet the simulation inserts single atoms. Equilibrium with X_2 fixes $\mu_X = \frac{1}{2}\mu_{X_2}$. Measured from separated atoms on the calculator's scale [Reuter–Scheffler]:

$$\mu_X(T, p) = \tfrac{1}{2}\left[\mu_{X_2}(T, p) - D_0\right] + e_{\text{atom}}, \tag{3}$$

where D_0 is the zero-point-corrected dissociation energy and μ_{X_2} is measured from the molecule's ground state:

$$\mu_{X_2} = k_B T \ln \frac{p\,\Lambda_{X_2}^3}{k_B T} - k_B T \ln(g_{\text{el}}\, q_{\text{rot}}\, q_{\text{vib}}), \quad q_{\text{rot}} = \frac{T}{\sigma\,\theta_{\text{rot}}},\ \theta_{\text{rot}} = \frac{\hbar^2}{2 I k_B},\ I = \frac{m_{X_2}}{4}\, r_e^2,\ q_{\text{vib}} = \frac{1}{1 - e^{-\theta_{\text{vib}}/T}}. \tag{4}$$

- **Gas-phase constants** (bond length r_e, vibrational wavenumber, symmetry number $\sigma = 2$, electronic degeneracy g_{el}, which is 3 for triplet O_2) are real spectroscopic constants. They are independent of the calculator.
- D_0 **and** e_{atom} must come from the calculator itself: `DiatomicReservoir.from_calculator(..., d0="self")` evaluates them with the potential the simulation uses. An approximate potential has its own energy scale, and mixing in an experimental D_0 would shift every μ by the potential's error. With `d0="self"`, $D_0 = D_e - \frac{1}{2}\hbar\omega$ uses the potential's own well depth and harmonic frequency, while q_{vib} uses the spectroscopic wavenumber.
- **Zero-point energy enters only the gas side.** μ contains the molecule's zero-point energy (about 0.13 eV per H atom for H_2), but the adsorbed atoms are classical. For light atoms the asymmetry is visible; compare with D_2 experiments or add harmonic zero-point corrections afterwards.

Figure 1 shows $\Delta\mu = \frac{1}{2}\mu_{X_2}$ for H and O. With the rigid-rotor, harmonic-oscillator model of Eq. (4) it agrees with the NIST-JANAF tables to within 7 meV per H_2 and 2 meV per O_2 at 300–1000 K. A unit test checks this (`tests/test_reservoir.py`).

1.4 Grand-canonical Monte Carlo moves

Each replica changes N and **R** with three moves; each one satisfies detailed balance with respect to Eq. (1).

Move	Proposal	Acceptance
Displacement	one atom by δ uniform in $[-\delta_{\max}, \delta_{\max}]^3$	$\min[1,\, e^{-\beta\Delta U}]$
Insertion	a new atom uniform in the insertion zone V_{ex}	$\min\left[1,\, \frac{V_{\text{ex}}}{(N_{\text{ex}}+1)\Lambda^3}\, e^{\beta(\mu-\Delta U)}\right]$
Deletion	one of the N_{ex} atoms inside V_{ex}	$\min\left[1,\, \frac{N_{\text{ex}}\Lambda^3}{V_{\text{ex}}}\, e^{-\beta(\mu+\Delta U)}\right]$

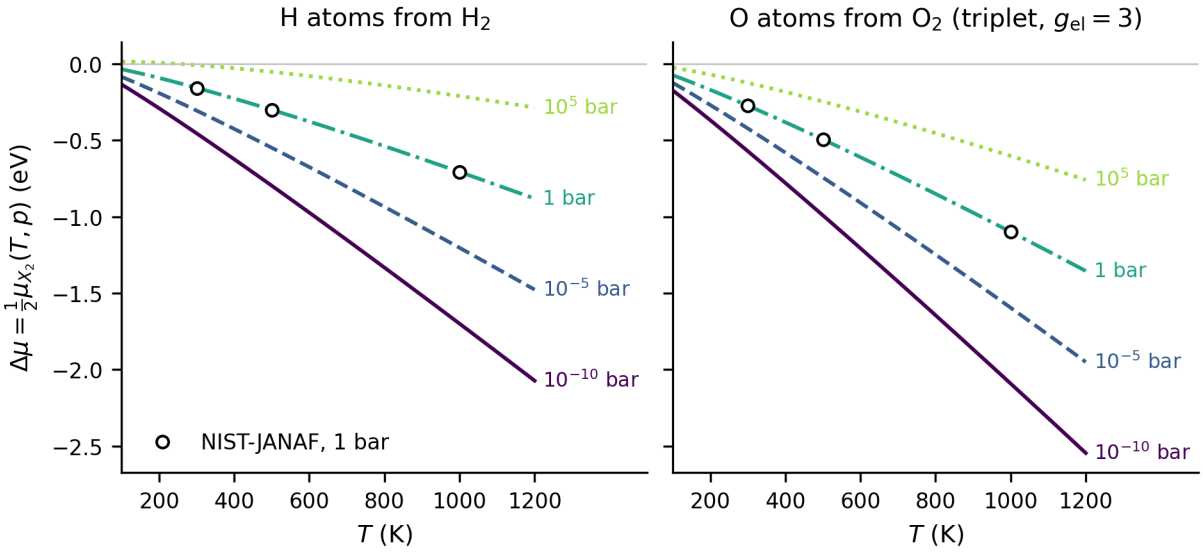


Figure 1. Chemical potential of an exchanged atom (per atom, measured from half the molecule's ground state) against T at four pressures, from `regc.diatomic_gas_mu_eV`. Circles: NIST-JANAF G – H(0) at 1 bar.

Insertions are restricted to an insertion zone V_{ex}: the space above a slab up to a chosen height, or a shell around a cluster. Deletions pick only among the N_{ex} atoms inside that zone. With this pairing the stationary distribution is still the full ensemble of Eq. (1) over the whole box (algorithm notes, §2.5). An ideal gas, for example, gives exactly $\langle N\rangle = V_{box}e^{\beta\mu}/\Lambda^3$ with a zone smaller than the box (`tests/test_paper_consistency.py`). This needs the diffusion to carry atoms across the zone boundary: an atom outside the zone can only be deleted after it has moved back in.

Insertion and deletion must be attempted with equal probability. The package refuses `p_insert != p_delete`. In an ideal-gas check, `p_insert = 0.4` and `p_delete = 0.1` gave ⟨N⟩ four times too large.

1.5 Replica exchange in (T, μ)

A single GCMC chain gets trapped at low T and between coverages separated by nucleation barriers. REGC runs an $L\times M$ grid of replicas and occasionally swaps the configurations of neighbouring grid points $a = (T_l, \mu_m)$ and $b = (T_{l'}, \mu_{m'})$. The swap is accepted with

$$P_{acc} = \min\left[1,\ \exp\Big((\beta_a - \beta_b)(U_a - U_b) - (\beta_a\mu_a - \beta_b\mu_b)(N_a - N_b) + (N_a - N_b)\ln\frac{\Lambda_a^3}{\Lambda_b^3}\Big)\right]. \tag{5}$$

Equation (5) follows from the product of two copies of Eq. (1). The algorithm notes, §3.1, explain the sign of the μ term, which is misprinted in [ZSG19]. Swaps go along T, along μ, and along the two diagonals (Figure 2). Each pattern pairs disjoint neighbours. The parity (offset 0 or 1) is drawn at random with the pattern, so every neighbour pair is reachable.

1.6 The REGC step

One REGC step [ZSG19, Fig. 1] is a grand-canonical move or a replica exchange, followed by a short canonical "diffusion" at fixed N that relaxes the structure. Here x_0 is the probability of the grand-canonical branch, as in the text of [ZSG19] ($x_0 = 0.99$ with swaps "every 100 steps"); their Fig. 1 labels the two branches the other way round.

The diffusion is one of four modes:

`diffusion`	What it does	Exact?
`mc_moves`	n single-atom displacement attempts ([ZSG19]'s LJ protocol)	yes

`diffusion`	What it does	Exact?
`mc`	n MC cycles, each atom attempted once per cycle on average	yes
`hmc`	n hybrid-MC trajectories (leapfrog + Metropolis on $H = U + K$)	yes, for any time step
`md`	n steps of thermostatted MD (Bussi or Langevin; natively in LAMMPS)	time-step bias

For MD, fresh Maxwell–Boltzmann velocities are drawn at the start of every segment. The thermostat counts 3N degrees of freedom, because frozen atoms and walls break momentum conservation. With LAMMPS's default count of $3N - 3$, 2-atom systems ran at 55 % of the target kinetic energy.

1.7 MBAR: from samples to every (T, μ)

Each sample n, recorded at any grid point, has a reduced potential at an arbitrary state (T, μ):

$$u(\mathbf{x}_n; T, \mu) = \beta U_n - \beta\mu N_n + N_n \ln \Lambda^3(T). \tag{6}$$

This is Eq. (1) without the state-independent $V^N/N!$. MBAR [Shirts & Chodera] solves self-consistently for the free energies f_k of the sampled states, and from them gives normalised weights at any target state:

$$w_n(T,\mu) = \frac{e^{\,f(T,\mu) - u(\mathbf{x}_n;T,\mu)}}{\sum_k N_k\, e^{f_k - u(\mathbf{x}_n;T_k,\mu_k)}}, \qquad \sum_n w_n = 1, \qquad \langle A\rangle_{T,\mu} = \sum_n w_n A_n, \tag{7}$$

where $f(T,\mu)$, the reduced free energy of the target state, is fixed by the normalisation.

- **Any function of the configuration** can be reweighted this way: N, the adsorbed count, coordination numbers, an RDF, U and U² for the heat capacity.
- **Phases** are labelled by N (or by another integer descriptor). The probability of phase i is $p_i = \langle \chi_i \rangle$, its free energy is $f_i = -k_B T \ln p_i$, and the stable phase is the most probable one [ZSG19, Eq. 8].
- **Reliability** is measured per target state by the Kish effective number of samples,

$$n_{\text{eff}}(T,\mu) = \Big(\sum_n w_n^2 \Big)^{-1}. \tag{8}$$

At a simulated state it is at least that state's own sample count, and more where neighbouring replicas overlap with it, because MBAR pools them. It approaches 1 where one sample dominates: extrapolation beyond the sampled range, or a gap between replicas that do not overlap. Correlated samples make the true count smaller by the statistical inefficiency. The package greys out and hatches cells with n_{eff} below a chosen threshold (Figures 5 and 10).

1.8 Validation on an exactly solvable case

With $U \equiv 0$ (an LJ potential with $\varepsilon = 0$), Eq. (1) is an ideal gas: $\langle N\rangle = V e^{\beta\mu}/\Lambda^3$ and $P(N)$ is Poisson. Figure 3 shows a run of the mixed sweep scheme (`ReplicaExchangeGCMC.run`, reduced units, $\Lambda^3 = 1$, 2×6 replicas, 1500 samples each): the direct averages and MBAR reweighting between and beyond the sampled μ, both against the exact result.

1.9 Heat capacity and phase boundaries

Coverage maps show where the most probable N changes, but not whether the change is a phase transition or a smooth crossover. [ZSG22] locate phase boundaries from the heat capacity

$$C_V(T,\mu) = \frac{\langle E^2\rangle - \langle E\rangle^2}{k_B T^2}, \qquad E = U + K. \tag{9}$$

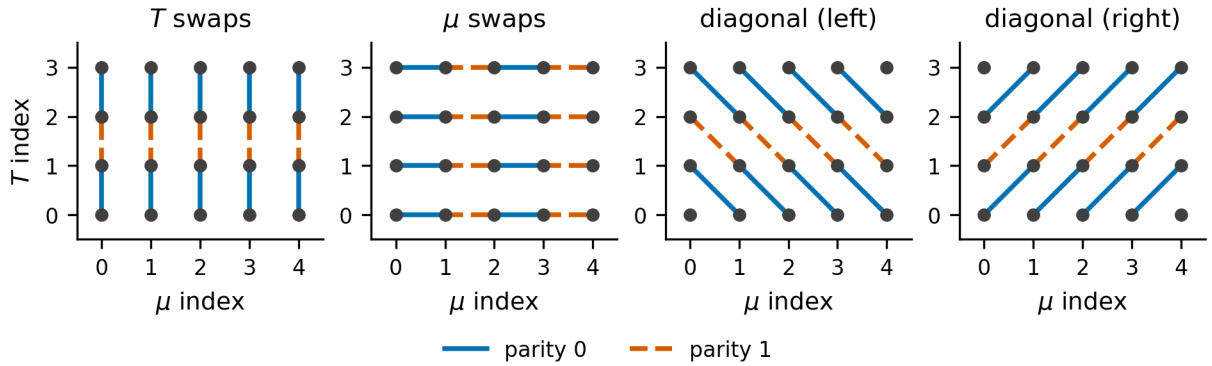


Figure 2. The four swap patterns of [ZSG19, Fig. 2] on a 4 × 5 grid, drawn from `ReplicaExchangeGCMC.swap_pairs`. Solid and dashed: the two parities.

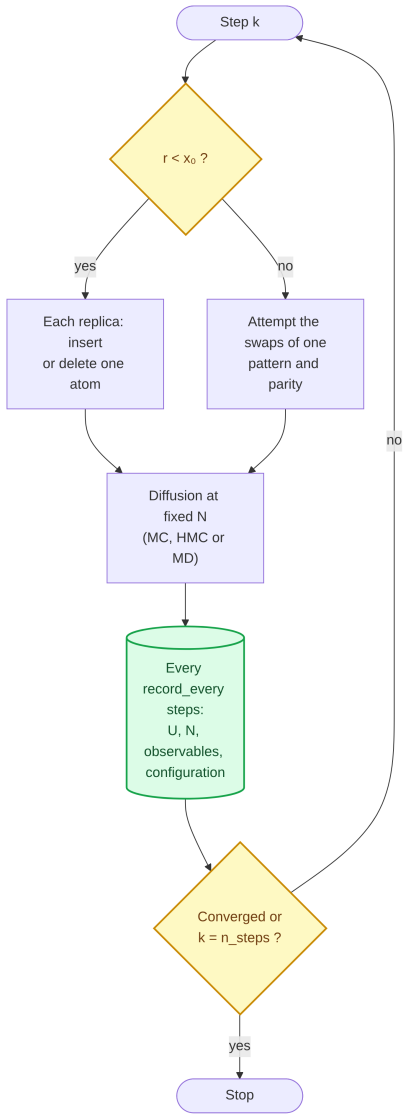


Diagram 1. One REGC step. With probability x0 every replica attempts an insertion or deletion, otherwise one swap pattern is attempted across the grid; then every replica diffuses at fixed N, a sample is recorded every record_every steps, and the run stops when converged or after n_steps.

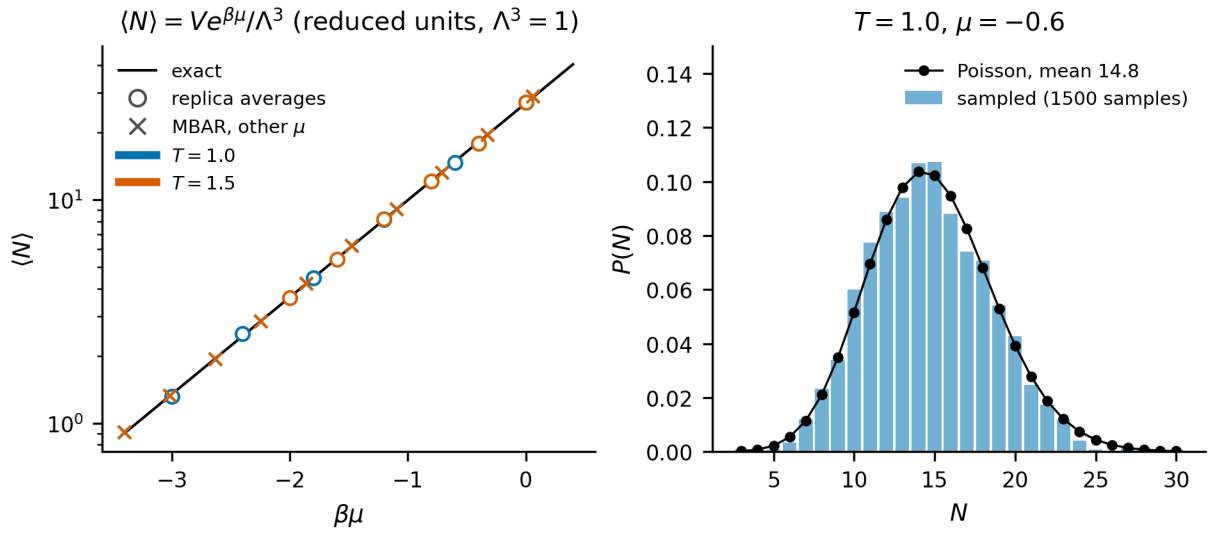


Figure 3. Ideal-gas validation ($V = 27$, reduced units). Left: $\langle N\rangle$ against $\beta\mu$ from the samples of each replica (circles) and from MBAR at other μ (crosses), with the exact law (line; both temperatures fall on it). Right: sampled $P(N)$ at one state (1500 samples) against the Poisson distribution with the exact mean. No error bars: successive samples are correlated, so the scatter of the points is the honest measure.

The kinetic energy is not sampled, but for n moving atoms (the mobile ones plus any displaceable substrate) it is known exactly given n: $\langle K \,|\, n\rangle = \frac{3}{2} n k_B T$ and $\mathrm{Var}(K \,|\, n) = \frac{3}{2} n (k_B T)^2$. With $x = U + \frac{3}{2} k_B T\, n$, this gives

$$\mathrm{Var}(E) = \mathrm{Var}(x) + \tfrac{3}{2}(k_B T)^2 \langle n\rangle. \tag{10}$$

Both moments come from Eq. (7). Differentiating $\langle E\rangle$ through the weights of Eq. (1), with $\Lambda^3 \propto T^{-3/2}$, shows that Eq. (9) is exactly $\partial\langle E\rangle/\partial T$ **at fixed** $\beta\mu$. It is not the heat capacity at fixed p or fixed N. A unit test checks this identity against a finite difference of the reweighted $\langle E\rangle$ (`tests/test_observables.py`). For an ideal gas, Eq. (10) with Poisson-distributed N gives $C_V/k_B = \frac{15}{4}\langle N\rangle$. The run of Figure 3 gives 55.1, against $\frac{15}{4} \cdot 27e^{-0.6} = 55.6$ exactly. (With the reduced-units convention $\Lambda^3 = 1$, Eq. (9) is still the [ZSG22] fluctuation formula, but no longer equals the derivative.)

C_V is extensive and includes the adsorption term (the energy per atom times $\partial\langle N\rangle/\partial T$). Values of 10^3–$10^4\, k_B$ for a few tens of adsorbates are therefore normal near a coverage step.

A first-order phase boundary shows up as a **ridge** of C_V: a maximum along μ (or p) at fixed T, and along T at fixed μ or p, whichever the mesh holds. `heat_capacity_ridges` finds these maxima, and [ZSG22] draw the boundaries through them. A change of coverage without a ridge is a smooth crossover; a ridge that fades out ends at a critical point.

Two effects produce ridges that are not phase boundaries:

- **Finite size.** In a small cell every $N \to N + 1$ step produces some C_V bump. A genuine first-order transition is a ridge that stays sharp and tall and grows with the cell, with a bimodal $P(N)$ across it.
- **Poor overlap.** C_V is a variance. Between two replicas whose samples barely overlap, MBAR rests on few effective samples and underestimates it. The result is peaks that sit *on* the sampled μ (or p) at every temperature (§4.1, Figure 7). `heat_capacity_ridges(..., ladder=...)` flags such points and warns. A real boundary moves with T, and stays where it is when the ladder is made denser.

1.10 Structural observables

The phase label N does not distinguish two structures with equal coverage, such as ordered and disordered dimers in [ZSG22]. Structural descriptors do. Each one is recorded with every sample and reweighted by Eq. (7):

Observable	Recorded per sample	Reported
`CoordinationNumber`	bonds n_b (centre–neighbour pairs with $r < r_c$), number of centres n_c	$\langle n_b \rangle / \langle n_c \rangle$
`CoordinationHistogram`	number of centres with CN = 0, 1, ..., $\mathrm{CN}_{\max}$	P(CN) of a centre atom ([ZSG22]'s H_{coord})
`RadialDistribution`	pair-distance histogram $H(r)$, n_a, n_b, V	$n(r) = \langle H(r) \rangle / (\langle n_a \rangle \Delta r)$; its integral to r_c is the coordination
`FunctionObservable`	anything `f(system)` returns	$\langle f \rangle$

The selection `'substrate'` means all slab or cluster atoms, frozen and displaceable; `'mobile'` means the exchanged atoms.

Ratios are formed after reweighting: ⟨bonds⟩/⟨centres⟩, not ⟨bonds/centres⟩. The latter weights a configuration with one centre as much as one with fifty, and is undefined with none. Centre atoms can be restricted to a height range, such as [ZSG22]'s 3 Å "surface region". Distances follow the cell's minimum-image convention.

1.11 Is the run converged?

[ZSG19] stop when there is "no change in the density of reduced-energy states ρ(U), with increasing simulation steps" (p. 6). `regc` compares the histograms of u (Eq. 6) of two **disjoint** consecutive blocks of samples. Comparing the samples at the previous check with *all* samples, which include them, gives a change that decays like 1/k after k checks whatever the data do. Figure 4 shows that comparison passing a series that drifts by ten standard deviations. Two blocks of n samples from one distribution still differ by about $\sqrt{\mathrm{bins}/\pi n}$ (the expected total-variation distance for equally populated bins), so the tolerance must lie above that noise floor. Here n counts *independent* samples, and the criterion takes the maximum over all replicas, which raises the floor further.

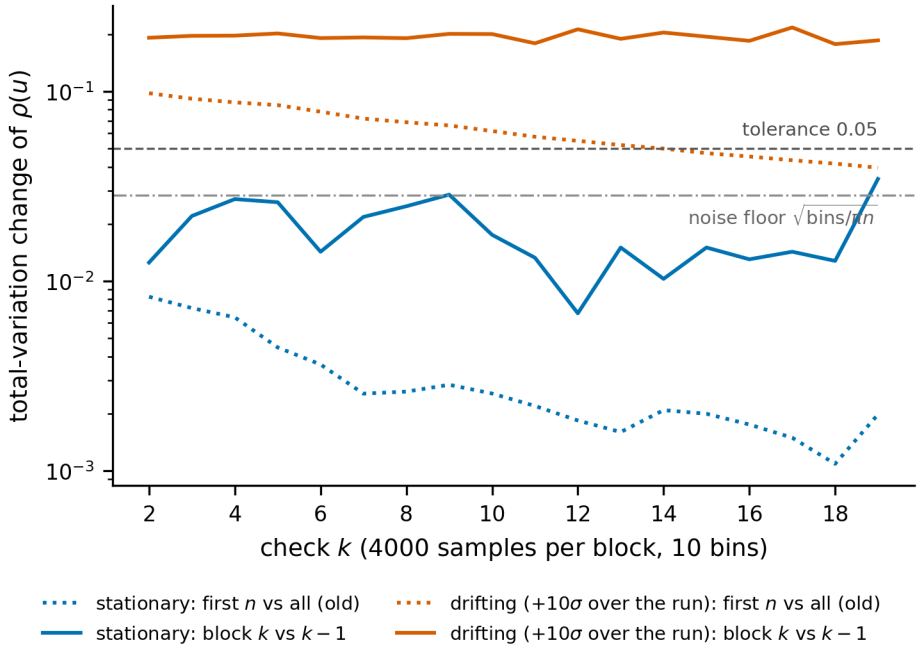


Figure 4. Convergence statistic for a stationary and a drifting synthetic series. Dotted: the nested comparison (decays regardless, and passes the drifting series). Solid: disjoint consecutive blocks, as `regc` now computes them.

A converged ρ(u) is necessary, not sufficient. Also check that the results do not change when you discard more of the start of the run (§8).

2. How the package is organised

2.1 Modules

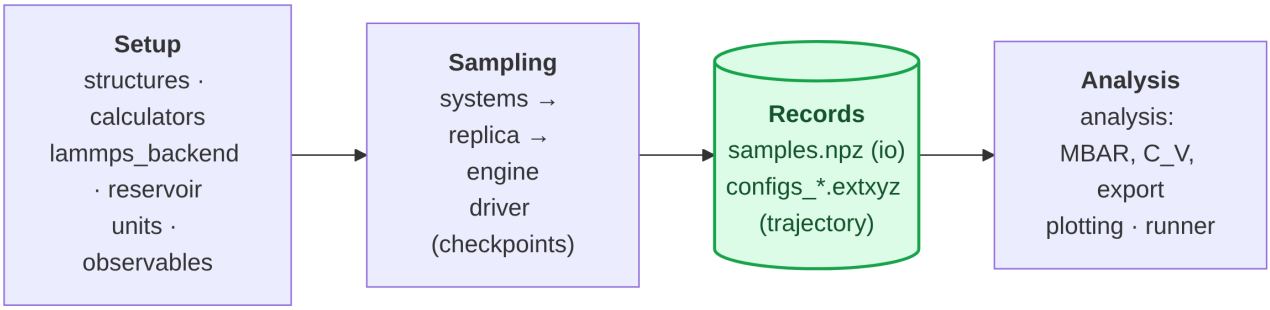


Diagram 2. Modules of the regc package. Setup modules build systems and reservoirs, the engine samples them using calculators and records observables and configurations, and the analysis modules turn the saved samples into maps, plots and exported structures.

The analysis also reads the reservoir, to convert μ to pressure.

Module	Main names	Needs
`systems`	`SlabSystem`, `ClusterSystem`, `BulkSystem`	numpy
`calculators`	`LJCalculator`, `PairLJCalculator`, `ASECalculatorAdapter`	numpy (ASE for the adapter)
`lammps_backend`	`LammpsCalculator`, `eam_pair_commands`, `lammps_ase_calculator`	LAMMPS Python module
`structures`	`slab_geometry_from_atoms`, `cluster_geometry_from_atoms`, `cubic_lattice_constant`	ASE
`reservoir`	`MonatomicIdealGasReservoir`, `DiatomicReservoir`	numpy
`engine`, `replica`	`ReplicaExchangeGCMC`, `Replica`	numpy
`observables`	`CoordinationNumber`, `CoordinationHistogram`, `RadialDistribution`, `FunctionObservable`	numpy
`driver`, `io`	`run_with_checkpoints`, `save_samples`, `load_samples`	numpy
`analysis`	`MBARSurfaceAnalysis`, `heat_capacity_ridges`, `export_phase_configurations`	pymbar, scipy
`plotting`	`plot_map`, `plot_phase_diagram`, `plot_radial_distribution`	matplotlib
`runner`	`regc-run config.yaml`	PyYAML

Optional dependencies are imported inside the functions that need them, so `import regc` works with numpy alone.

2.2 Parallel execution

Between two replica exchanges the replicas are independent. The engine sends each replica a whole block of steps, to a pool of worker processes when `n_workers > 1`. Observables are evaluated in the workers; the main process collects the samples and writes the configuration files.

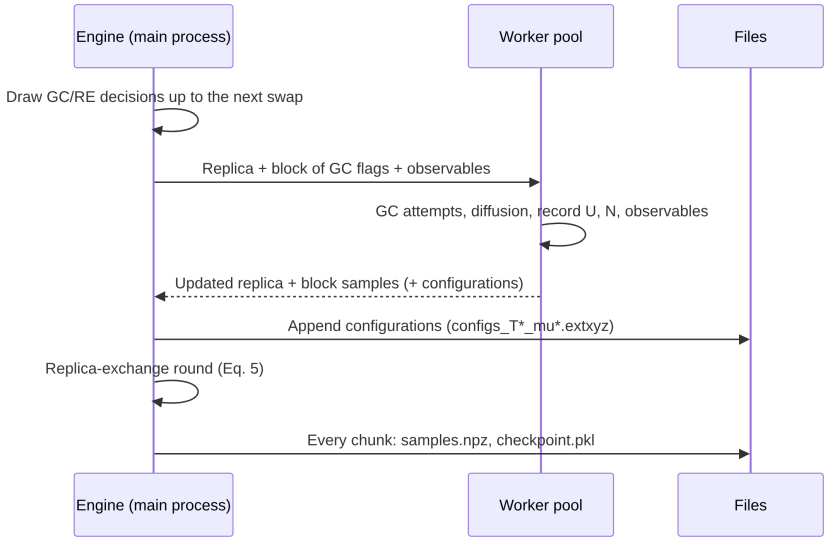


Diagram 3. One block of steps in a parallel run. The engine draws the random decisions of a block, ships each replica to a worker that runs the block and evaluates observables, receives samples and configurations back, writes the files, and then performs the replica exchange in the main process.

Each replica owns its random stream and it travels with the replica, so runs with 1 and with many workers give bit-identical samples.

---

3. A calculation step by step

A real-units calculation has six steps. The snippets follow H on an Fe surface with an EAM potential in LAMMPS (`examples/13_fe_h_eam_surface.py` is the complete script).

Step 1 — the structure

```
from ase.build import bcc100
from regc import cubic_lattice_constant, eam_pair_commands, lammps_ase_calculator, \
    slab_geometry_from_atoms

pair = eam_pair_commands("FeH_Wen.eam.fs", ["Fe", "H"])
ase_calc = lammps_ase_calculator(pair, ["Fe", "H"])          # the same potential, in ASE
a = cubic_lattice_constant(ase_calc, "Fe", "bcc")              # the potential's own
 ↪ lattice constant
geometry = slab_geometry_from_atoms(bcc100("Fe", size=(3, 3, 6), a=a, vacuum=None),
                                    n_frozen_layers=3, headroom=6.0)
print(geometry.summary())
```

- **Frozen and free layers:** the bottom `n_frozen_layers` stay frozen; the layers above them relax and vibrate.
- **Coordinates:** the top frozen layer sits at z = 0, and `geometry.top_z` is the height of the topmost atom.
- **Headroom:** the gas atoms live in the `headroom` above the slab.
- **Clusters:** `cluster_geometry_from_atoms` does the same for a nanoparticle in a non-periodic box.

Step 2 — the calculator and the reservoir

```
from regc import DiatomicReservoir, LammpsCalculator, kelvin_to_eV

h2 = DiatomicReservoir.from_calculator(ase_calc, "H")        # D0 and e_atom from the
 ↪  potential
print(h2.calculator_info)                                     # check r0, De, D0 against
 ↪  experiment

def make_system(rng):
    return geometry.make_system(LammpsCalculator(pair, ["Fe", "H"]), "H",
                                insertion_z_range=(geometry.top_z - 0.5, geometry.top_z +
 ↪  4.0))
```

Before a long run, check that the potential's H_2 molecule is reasonable (bond length, D_0). Also check the adsorption energy of a few sites with `insertion_energy_scan`.

Step 3 — the replica ladder

```
T_K = [300, 400, 500, 650, 800]
p_bar = [1e-8, 1e-6, 1e-4, 1e-2, 1, 1e2]
mus = h2.mu_grid(T_K, p_bar, unit="bar")                      # shape (5, 6), Eq. (3)
```

- **Spacing:** neighbouring replicas must overlap in energy and N. Aim for 20–40 % swap acceptance (§8).
- **Pressure grid:** a pressure grid gives a μ grid that is different at every T. A μ grid shared by all temperatures is also allowed.

Step 4 — observables and structures to keep

```
from regc import CoordinationNumber, RadialDistribution

observables = [
    CoordinationNumber("cn_HFe", r_cut=2.2, center="mobile", neighbor="substrate"),
    RadialDistribution("rdf_HFe", r_max=4.0, n_bins=80, a="mobile", b="substrate"),
]
```

Step 5 — sampling

```
from regc import ReplicaExchangeGCMC, run_with_checkpoints

with ReplicaExchangeGCMC([kelvin_to_eV(T) for T in T_K], mus, make_system,
                         lambda_cubed_of_T=h2.lambda_cubed_of_T(), n_workers=30,
                         adsorbed_cutoff=2.2, observables=observables,
                        configurations_dir="run/configurations", configurations_every=2)
 ↪  as sim:
    status = run_with_checkpoints(sim, "run", n_steps=20000, chunk=1000, x0=0.9,
                                  diffusion="md", diffusion_steps=40, record_every=10,
                                  timestep_fs=0.5, taut_fs=20.0)
```

- **Λ^3:** `lambda_cubed_of_T` must be the Λ^3 of the **exchanged atom**, the one the reservoir defines.
- **Chunks:** the run is saved every `chunk` steps and resumes after a crash or a wall-time limit (§7).

Step 6 — analysis

```
import numpy as np
from regc import MBARSurfaceAnalysis, heat_capacity_ridges, load_samples, plot_map

a = MBARSurfaceAnalysis(load_samples("run/samples.npz", discard=0.2),
                        reservoir=h2, pressure_unit="bar")
T_grid_K = np.linspace(300, 800, 51)
T_grid = np.array([kelvin_to_eV(T) for T in T_grid_K])
p_grid = np.logspace(-8, 2, 81)
theta = a.phase_diagram_Tp(T_grid, p_grid, area=geometry.n_surface, observable="N_ads")
ess = a.effective_sample_size(T_grid, p_grid, pressure=True)
cv = a.heat_capacity(T_grid, p_grid, pressure=True)
ridges = heat_capacity_ridges(T_grid_K, p_grid, cv, valid=ess >= 20)
plot_map(T_grid_K, p_grid, theta, "coverage.png", valid=ess >= 20,
 ↪  points=ridges["points"],
        colorbar_label="H per surface Fe", xlabel="p(H2) (bar)", ylabel="T (K)")
```

The analysis section (§5) covers the remaining tools.

---

4. Worked examples

Example	System	Shows
`01_validate_calculator.py`	LJ	the native LJ against a brute-force sum
`02–04`	LJ bulk and surface	parallel runs, frozen substrate, a first phase diagram
`05_emt_metal_surface.py`	H on Au(111), EMT	real units and Λ^3 of H, with μ set directly
`10_hmc_demo.py`, `11_...`	LJ with a relaxing substrate	HMC and MD diffusion
`12_paper_lj_surface.py`	[ZSG19] §III A	the paper's LJ surface, MBAR against aiAT (its docstring: about 80 min on 12 cores)
`13_fe_h_eam_surface.py`	H on Fe, EAM in LAMMPS	a production workflow with checks and a CLI
`14_observables_heat_capacity.py`	[ZSG19] LJ surface	observables, C_V, structures of each phase (7–8 min on 12 cores, §4.1)
`pd111_o_grace/`, `pt55_o_grace/`	O on Pd(111), O on Pt_{55}; GRACE MLIP	config-file runs on a cluster

4.1 The LJ surface with observables, heat capacity and structures

This is `examples/14_observables_heat_capacity.py`.

The system [ZSG19, §III A]. A frozen two-layer fcc(111) slab of 18 A atoms in a 3 × 3 hexagonal cell (lattice vector 11.489 Å). B atoms adsorb from an ideal gas into a 48 Å zone above the top A layer, which sits at z = 0. A hard wall there keeps B out of the slab, as in the paper's setup; without it, B slips through the hollows and is absorbed between the A layers (algorithm notes, §7). Interactions are LJ: $\varepsilon_{AB} = 0.66$ eV, $\sigma_{AB} = 1.91$ Å, $\varepsilon_{BB} = 0.01$ eV, $\sigma_{BB} = 1.2$ Å. The mass of B, 16.48 amu, is not stated in [ZSG19]; it is inferred from their Fig. 3 caption (algorithm notes, §7).

Two runs. Both use six temperatures (200–650 K), 2×10^4 REGC steps with $x_0 = 0.99$, ten MC moves of diffusion per step, and a sample every 50 steps; after discarding the first 20 %, 320 samples per replica remain. Each takes 7–8 minutes on 12 cores.

Run	μ ladder	Swap acceptance (T / μ / diagonals)
coarse	11 values, −2.4 to −0.9 eV in steps of 0.15 eV	0.53 / 0.23 / 0.21–0.25
dense	11 values, −1.65 to −1.25 eV in steps of 0.04 eV	0.32 / 0.40 / 0.27–0.30

Every sample records four observables:

- the B–A coordination (r_c = 2.8 Å, beyond the 2.14 Å pair minimum);
- the B–B coordination histogram;
- the B–B and B–A RDFs.

Every sample's configuration is stored as well.

Both runs are illustrations, a sixth of the paper's length. They are not converged enough for a quantitative comparison with [ZSG19, Fig. 3], whose most prominent phases are N_B = 0, 18, 45, 59 and 85. What they do show well is how to read, and how not to read, the maps.

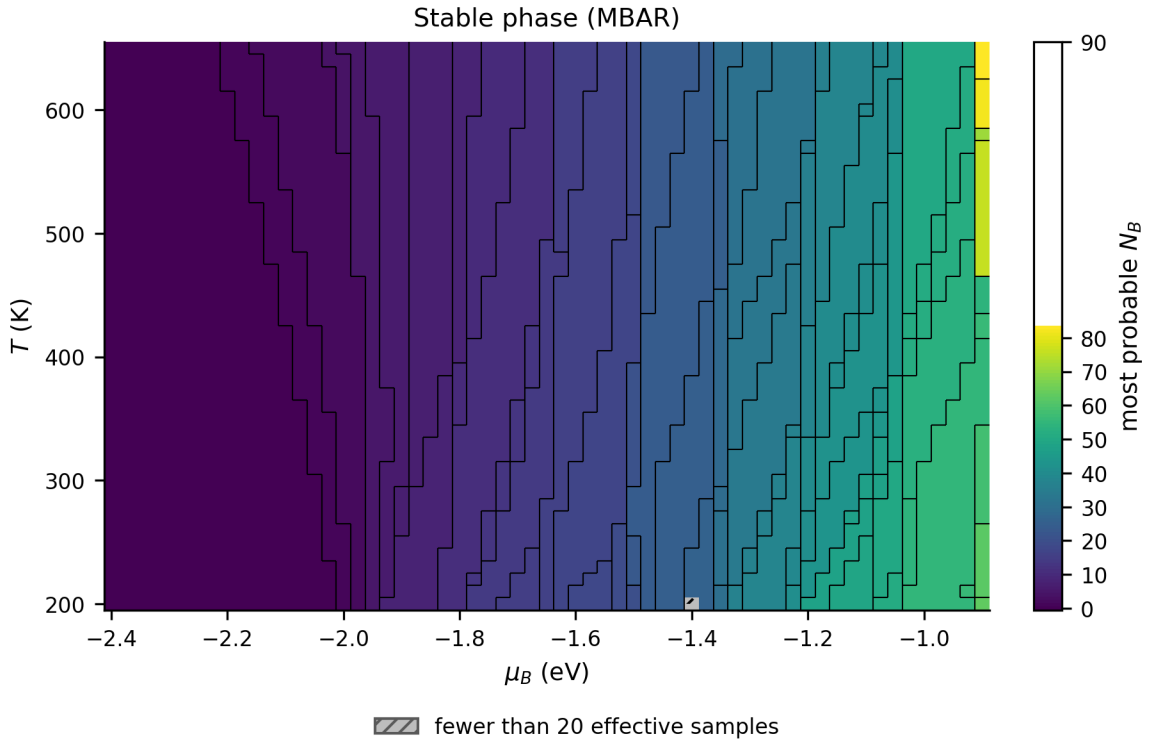


Figure 5. Coarse run: most probable N_B on a 46 × 61 (T, μ) mesh. Black lines separate cells of different N_B. Hatched: fewer than 20 effective samples (one cell). The 56 distinct values change almost continuously, so colours are continuous.

The heat capacity needs overlap. In the coarse run, 83 % of the C_V peaks of Figure 6 lie on one of the 11 sampled μ, at every temperature. `heat_capacity_ridges(..., ladder=...)` flags this and warns. A phase boundary that does not move with T would need equal entropies on both sides, so vertical columns on the ladder are suspect from the start. Figure 7 tests them by repeating the run with a ladder almost four times denser over −1.65 to −1.25 eV.

- **Coarse ladder.** At 290 K, the two tallest peaks sit exactly on the sampled −1.50 and −1.35 eV.
- **Dense ladder.** Those peaks become broad, lower features. Between two replicas that barely overlap, MBAR's variance estimate rests on a few samples and collapses, which is what produced the peaks.

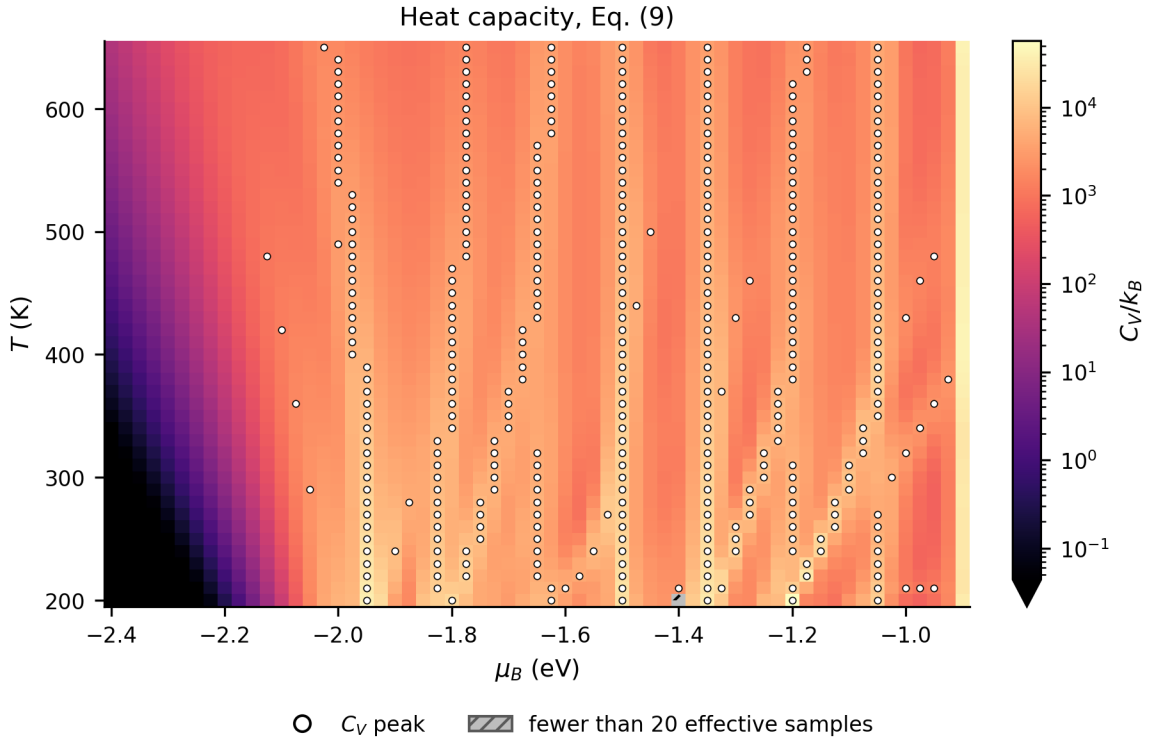


Figure 6. Coarse run: heat capacity C_V/k_B (Eq. 9, log scale; the colourbar arrow marks values below the 2nd percentile) with its peaks along μ and along T (white dots, prominence 0.3). Most peaks form vertical columns on the sampled μ: see Figure 7.

- **What survives.** The dense map still has short vertical streaks on its own ladder; 2×10^4 steps are not enough. It also has ridges that move with T at low temperature, the behaviour expected of real boundaries. Only a longer run, or a second ladder offset by half a step, can separate the two.

The same goes for the coverage. The coarse run's ⟨N_B⟩ along the 290 K isotherm rises in steps whose edges lie on the sampled μ. With the dense ladder it rises smoothly (Figure 8), and $P(N_B)$ stays broad and single-peaked as it shifts. Over this window, at 290 K and this cell size, adsorption is a continuous filling, not a sequence of first-order transitions.

Structures of the phases. `export_phase_configurations` labels every mesh cell with its most probable N_B and groups the cells into connected regions. For each region it writes the stored configuration with the largest MBAR weight, among those with that N_B, at the cell where that N_B is most probable. Figure 12 shows four of them; the extxyz files open in OVITO, ASE or pySNOW.

4.2 H on Fe with an EAM in LAMMPS

`examples/13_fe_h_eam_surface.py` is the production version of §3. It has two extra modes:

- `--check` verifies the setup before any sampling: lattice constant, the H_2 molecule, site energies, and energy bookkeeping over one REGC step.
- `--analyze-only` re-analyses a finished run.

MD diffusion runs natively in LAMMPS rather than through ASE's MD, which the algorithm notes (§2.2) report as about 10^3 times faster for an EAM slab.

LAMMPS's per-run setup dominates the cost of an energy call for a slab this size, so the calculator skips it when only positions change. It sends the new positions and runs one step without an integrator, and it reuses the energy of the configuration before a move and the end energy of each MD segment. Every energy still agrees with a full setup to round-off. Measured on a 60-atom Fe(100) + H slab on one core:

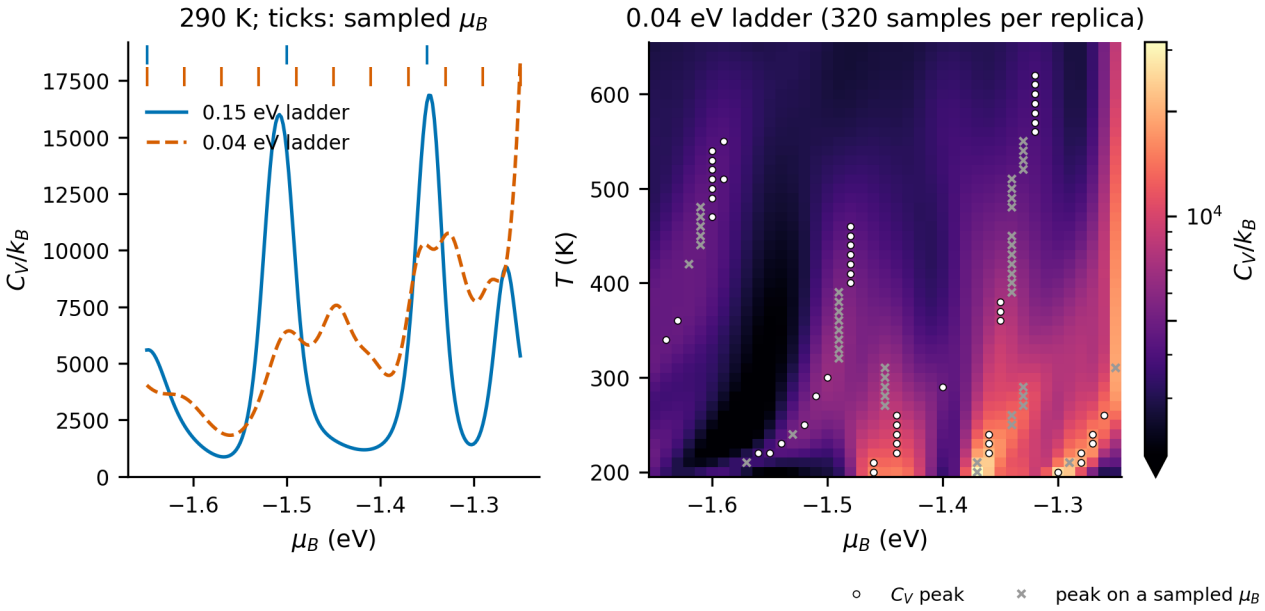


Figure 7. The overlap artefact. Left: C_V at 290 K from the coarse (solid) and the dense (dashed) ladder; ticks at the top mark each run's sampled μ_B. Right: dense-run C_V map with its peaks; grey crosses are peaks within 0.01 eV of a sampled μ_B (the mesh spacing, so this flag is coarse here).

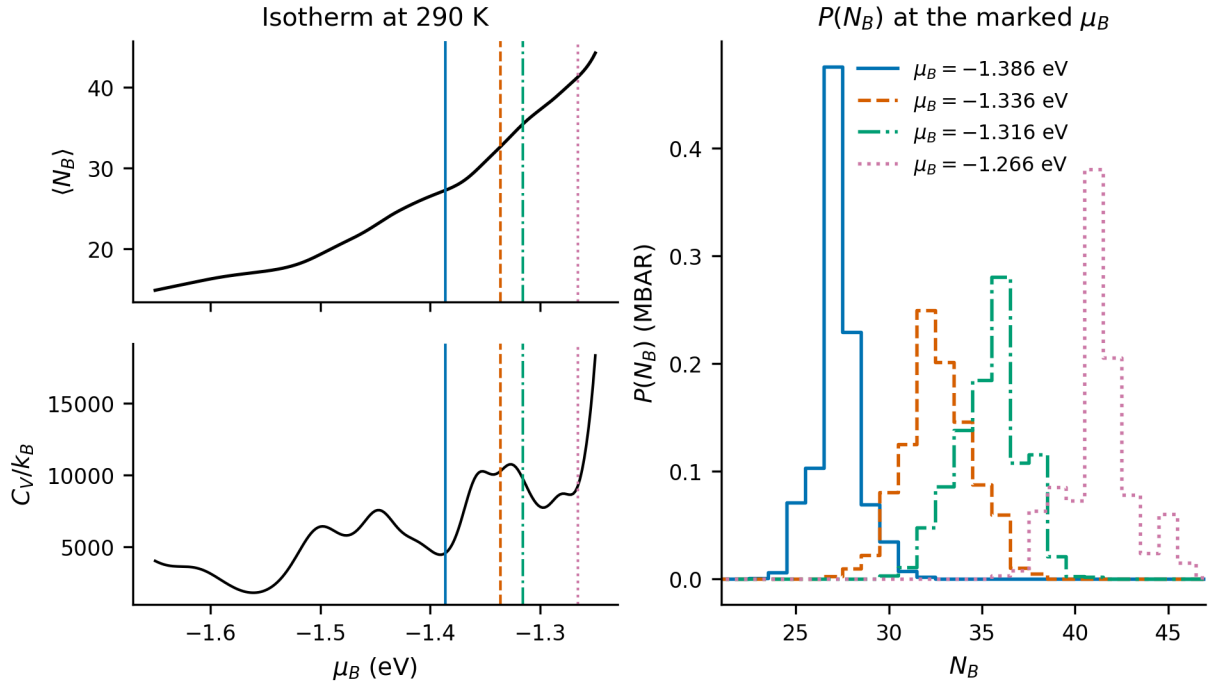


Figure 8. Dense run, 290 K. Top left: $\langle N_B \rangle$; bottom left: C_V; vertical lines mark the μ_B of the right panel. Right: reweighted $P(N_B)$ at those μ_B: single, broad peaks moving from $N_B \approx 27$ to ≈ 41.

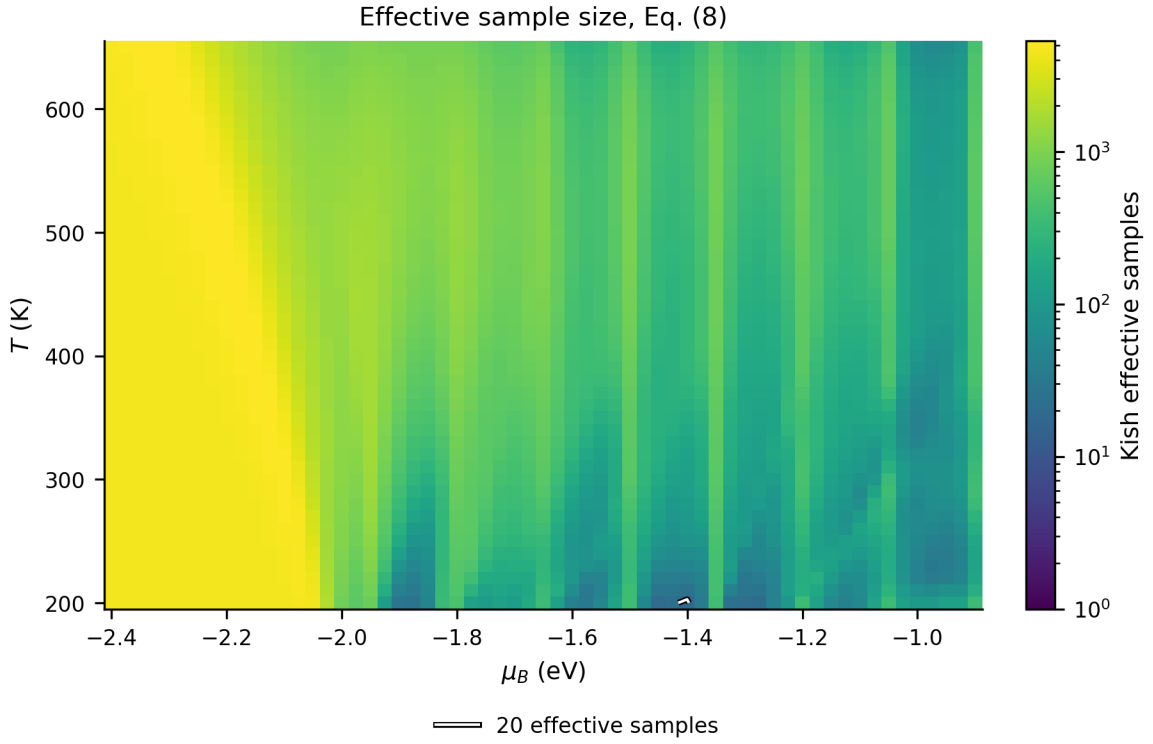


Figure 9. Coarse run: Kish effective sample size (Eq. 8). It is largest on the empty surface (μ < −2.05 eV), where samples from all replicas agree, and lowest between sampled μ at low T. The white line marks 20.

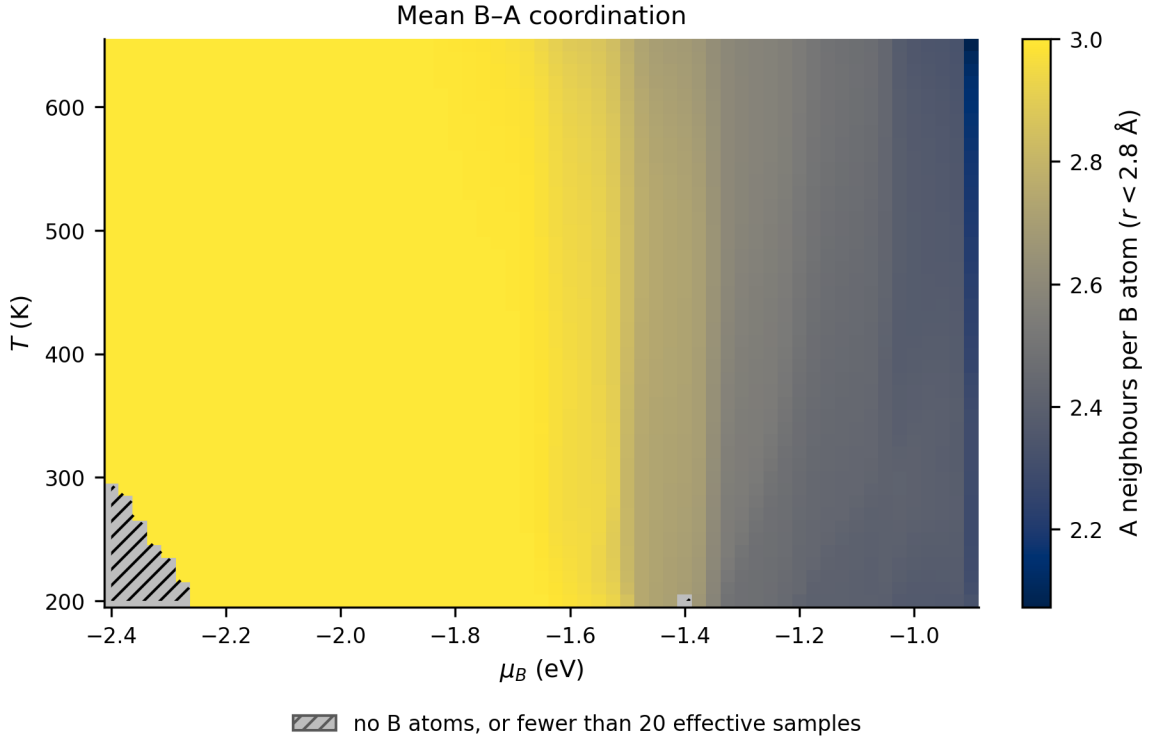


Figure 10. Coarse run: mean number of A neighbours of a B atom, ⟨bonds⟩/⟨centres⟩. It is 3 when B occupies the three-fold hollows of the first layer and falls as the second layer, farther from the A atoms, fills. Hatched at low μ and T: no B atoms, so no average.

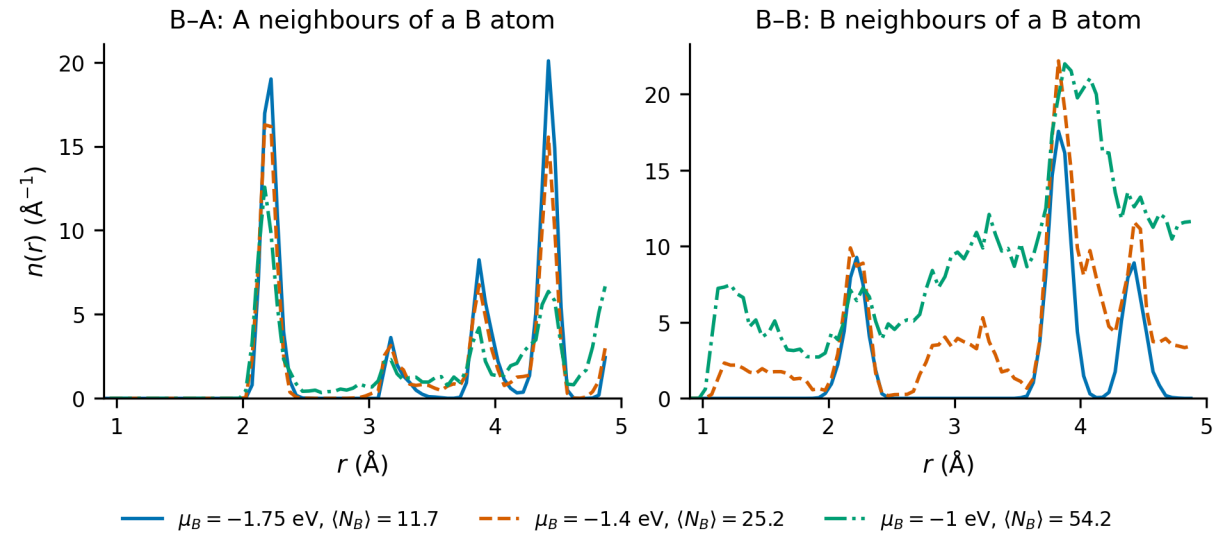


Figure 11. Coarse run: reweighted radial distributions at 290 K, per centre atom (the integral to r_c is the coordination). Left: B–A, with its first shell at the hollow-site distance $3.83/\sqrt{3} = 2.21$ Å, just beyond the 2.14 Å pair minimum. Right: B–B, building up as the layer fills.

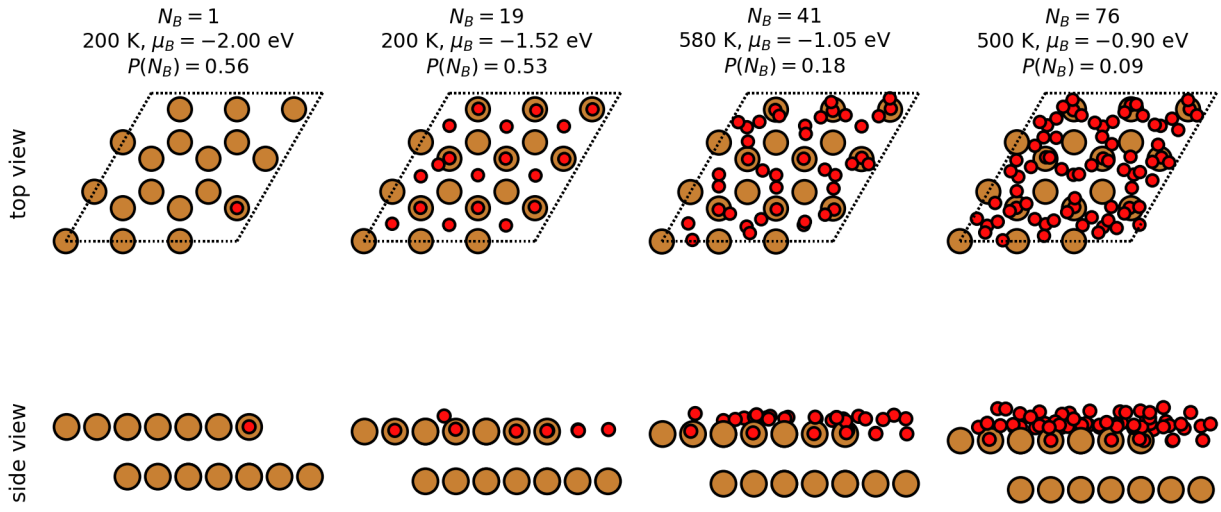


Figure 12. Four exported structures from the coarse run (top and side views; A copper-coloured, B red). Each title gives N_B, the state it was taken at and $P(N_B)$ there. First-layer B atoms sit in the plane of the top A layer, held there by the hard wall at z = 0: the hollow-site distance, 2.21 Å, exceeds the 2.14 Å pair minimum.

Path	Full setup every call	Fast path	Speed-up
MC displacement move	3.62 ms	0.27 ms	13×
GC attempt (count changes: full setup needed)	3.44 ms	2.21 ms	1.6×
REGC step, 10 MC moves per replica (2 × 2 replicas)	177 ms	31.6 ms	5.6×
REGC step, 40 MD steps per replica	64.6 ms	39.0 ms	1.7×

`REGC_LAMMPS_FAST=0` restores the full path for checks.

4.3 O on Pd(111) with a foundation MLIP (GRACE)

`examples/pd111_o_grace/` runs from a YAML file (§6) with a GRACE foundation model in LAMMPS (`lammps_name: grace`; the model is chosen with `GRACE_MODEL` in `run.sh`). The O_2 reservoir takes D_0 from the model (`d0: self`) and includes O_2's triplet ground state (`electronic_degeneracy: 3`, the default for O). The `run.sh` script submits to Slurm with automatic resubmission (§7).

---

5. Analysis in depth

5.1 The main tools

Task	Call
Coverage map	`phase_diagram_Tp(T, p, area, observable)` / `phase_diagram(T, mu, ...)`
Stable phase	`most_probable_N_Tp(T, p)` / `most_probable_N(T, mu)`
One state	`mean_N(T, mu)`, `pN(T, mu)` (reweighted P(N))
Reliability	`effective_sample_size(T, grid, pressure)`
Heat capacity	`heat_capacity(T, grid, pressure, kinetic=True)`, `heat_capacity_ridges(...)`
Observables	`observable_map`, `mean_coordination`, `coordination_distribution`, `radial_distribution`
aiAT comparison	`lowest_energy_N(mu)`, `lowest_energy_N_Tp(T, p)`
2D condensation	`locate_coexistence_p(T, bracket)`, `coexistence_curve_Tp(T, bracket)`
Structures	`representative_samples(...)`, `export_phase_configurations(...)`

Temperatures are always k_BT in the energy unit of U (eV in real units). Pressures are in the `pressure_unit` given to `MBARSurfaceAnalysis`.

5.2 What to plot, and how

The plotting helpers follow three rules:

- **Unreliable cells are shown, not hidden.** Cells with few effective samples are grey and named in the legend.

- **Colormaps match the data.** Integer phase labels get one colour per value. Continuous quantities use perceptually uniform maps (viridis, cividis, magma). C_V uses a logarithmic scale clipped at its 2nd percentile, and the colourbar arrow marks the clip.
- **Pressure is plotted as $\log_{10}$ p** on a uniform axis.

```
plot_map(T_K, p, most, "phases.png", valid=ess >= 20, discrete=True,
         points=ridges["points"], colorbar_label="most probable N")
```

5.3 The structures behind the diagram

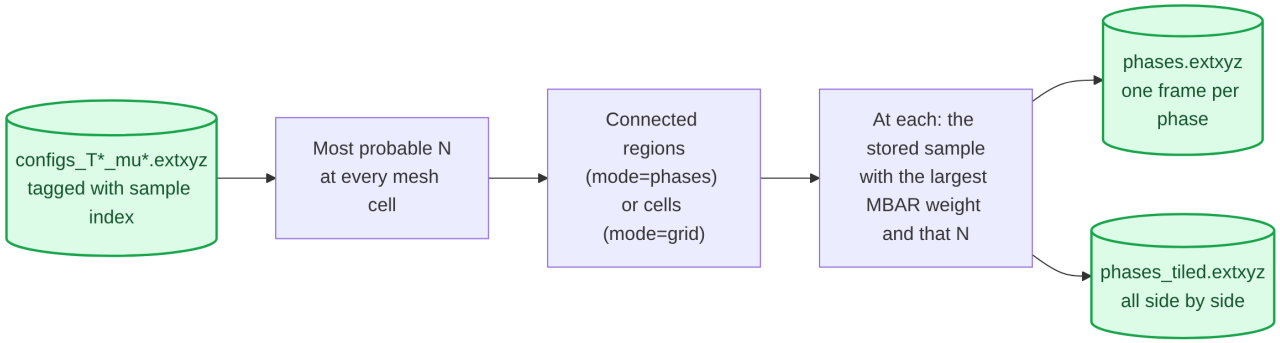


Diagram 4. From stored configurations to phase structures. Stored configurations are tagged with their sample index; MBAR weights at each mesh cell pick the most probable sample with the phase's N, which is read back from its file and written to an extxyz file, optionally tiled into one visual phase diagram.

Each exported frame records its state:

- the phase label and its probability there (`P_phase`);
- the effective sample size;
- the sample's weight and origin (`source_T_index`, `source_mu_index`, `source_sample`).

Each atom has a `group` column: 0 frozen substrate, 1 displaceable substrate, 2 mobile. The tiled file adds `tile_row` and `tile_col`, so a viewer can colour or label the tiles.

5.4 Reliability: correlation, overlap, error bars, mixing

A map from MBAR always looks smooth. `regc.diagnostics` answers four questions that decide how much of it to believe; `regc-run` runs the first three automatically and prints warnings.

Question	Tool	What to look for
How many independent samples?	`replica_inefficiencies` (uses `statistical_inefficiency`); `MBARSurfaceAnalysis(..., subsample="g")`	n/g per replica, with $g = 1 + 2\sum_t \rho(t)$ for N, U and u; fewer than about 20 makes averages, and especially C_V, unreliable. Thinning to every g-th sample makes pymbar's analytical errors honest, at the price of noisier estimates
Do neighbouring replicas overlap?	`neighbour_overlap(analysis)`	the MBAR overlap of each pair of grid neighbours; below about 0.03 [Klimovich] the states between them rest on few samples

Question	Tool	What to look for
How large is the statistical error?	`bootstrap(analysis, statistic, n_boot)` → `BootstrapResult`	percentile intervals (`interval`), standard errors (`std`), and for an integer map the fraction of resamples that agree with it (`agreement`)
Is a C_V peak real?	`ridge_support(ridges, cv_replicates)`, plus `heat_capacity_ridges(..., ladder=...)`	support close to 1, and a peak that is not on a sampled value

How the bootstrap works.

- **Blocks.** Each replica's samples are resampled in blocks of consecutive samples about $2g$ long, so the correlation inside a block is kept. Each replica is resampled independently, which ignores the weaker correlation that swaps create between replicas.
- **MBAR is re-solved for every resample.** It starts from the full-data free energies and uses a Newton solver that agrees with pymbar to round-off, at about a second per resample for 66 states and 21 000 samples.
- **Any statistic.** The statistic is any function of the analysis, such as a C_V map, a coverage map or the most probable N.

On the ideal gas, the 90 % intervals contain the exact $\langle N \rangle$ in at least 11 of 16 independent runs; this is a unit test.

One resample, step by step. The bootstrap [Künsch] imitates "run the simulation again" with the samples already in hand, and repeats the whole analysis on each imitation. In short, then step by step:

0. **Block length, once.** Each replica's statistical inefficiency g (the largest of those of N, U and u; §1.11) sets a block length $b = \lceil 2g \rceil$, capped at a quarter of its samples.
1. **Redraw.** For each replica, $\lceil n/b \rceil$ block starts are drawn uniformly in $[0, n-b]$ with replacement; the blocks of consecutive samples are concatenated and cut to n. Some stretches of the run appear twice, some not at all; correlation inside a block is kept. Replicas are redrawn independently: the correlation that replica exchange creates between neighbouring replicas is ignored.
2. **Copy.** Every per-sample column (U, N, N_ads, N_s, observables, energies under other potentials) is copied at those indices into a new sample set with the same ladder and metadata.
3. **Re-solve MBAR.** The resampled set gets its own free energies f_k from the MBAR equations $\sum_n W_{kn} = N_k$ (§1.7). The package's Newton solver starts from the full data's f_k and normally needs 4–5 steps: about 0.1 s for 32 states and 38 400 samples.
4. **Denominators.** For every sample it stores $\ln \sum_k N_k e^{f_k - u_k(x_n)}$. The weight at any target state is then one vector operation.
5. **Statistic.** The function you passed (a coverage map, a C_V map, the most probable N) is evaluated on the resampled analysis exactly as on the full data.
6. **Summary.** Over the `n_boot` replicates: `std()` (standard error), `interval(level)` (percentiles; with 40 resamples the 5th percentile is the second- or third-smallest value, so use a few hundred for quoted 95 % intervals), `agreement()` (integer labels).

Why some resamples took minutes. In the Fe–H benchmark of §5.6, about one resample in eight took 175–185 s instead of 0.1 s. Instrumenting the solver on a replay of those resamples showed the cause: the Newton iterations reached a gradient of a few 10^{-9} (the weight sums matched the sample counts to about 10^{-4} of a sample) and then could not go further. The decrease of the objective left along the Newton direction was below its floating-point resolution, so the line search halved the step about 25 times until round-off noise let a step of about 10^{-7} pass, 500 times over (the iteration cap). The solver now stops when only such negligible steps pass and the gradient is already below 10^{-6}. On the stalled resample the result is unchanged to 2.7×10^{-13} kT (the coverage map to 7×10^{-15}), the solve takes 1.2 s, and a 40-resample bootstrap of that run takes 58 s instead of up to 16 min. A unit test forces the situation with an unreachable target.

Comparing two estimates. To compare two maps with their own error bars (two runs, or a reweighted map and a direct one), use $z = (a-b)/\sqrt{\sigma_a^2 + \sigma_b^2}$ per cell: with correct errors and no bias, about 95 % of cells have $|z| < 2$. Checking whether

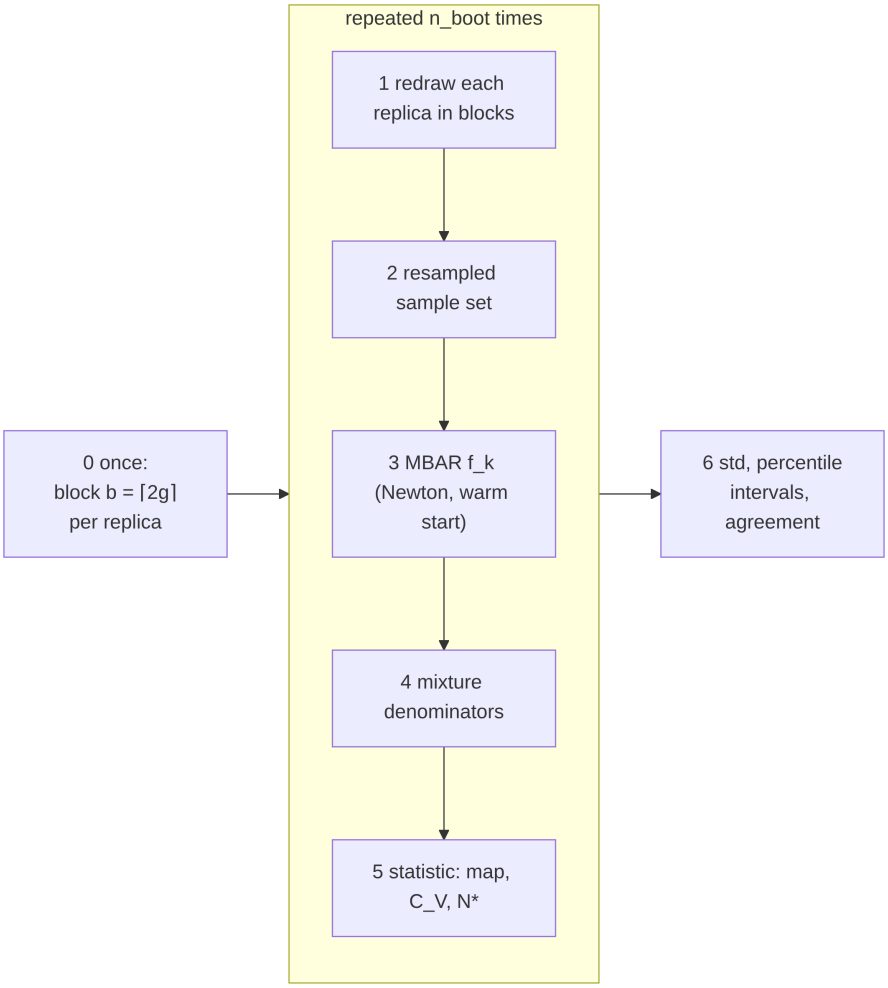


Diagram 5. One bootstrap resample. Block lengths are fixed once from each replica's statistical inefficiency; each resample redraws every replica's samples in blocks, builds a resampled sample set, re-solves the MBAR free energies by warm-started Newton, rebuilds the reweighting denominators and recomputes the statistic; the replicates give standard errors, percentile intervals and agreement.

b lies in a's 90 % band is a weaker test than it looks. Even for two perfect estimates with equal errors it succeeds in only about 75 % of cells, because the band ignores b's own error.

Mixing. The engine also counts swaps per neighbouring pair (`pair_acceptance`) and follows every configuration as a walker across the grid (`round_trip_counts`). A pattern's average acceptance can hide one pair that never swaps and cuts the ladder in two. A walker that never completes a round trip in T has not carried a low-temperature structure through the hot replicas and back. The driver writes both to `mixing.json` after every chunk.

What the LJ runs of §4.1 show (Figure 13).

- **Correlation.** The statistical inefficiency is about 43 in every replica. Each replica's 320 samples hold about 7 independent ones, which is far below 20: these runs are illustrations, as §4.1 says.
- **Overlap.** On the coarse ladder, 118 of 215 neighbouring pairs overlap less than 0.03. On the dense ladder, 28 of 215 do.
- **Bias versus noise.** The coarse run's C_V peaks on the sampled −1.50 and −1.35 eV have bootstrap support 1.00 and 0.98 (Figure 13; the one on −1.95 eV, outside the figure's window, has 1.00). They are reproducible, yet they are the overlap artefact of §4.1. The bootstrap measures variance, and that artefact is a bias, the same in every resample. Error bars and the overlap and ladder checks are therefore complementary; use both.
- **The dense run is not clean either.** Its broad, low peak on the sampled −1.45 eV also has support 1.00, so a run this short cannot settle that feature; its peak near −1.33 eV is supported by fewer than half of the resamples.

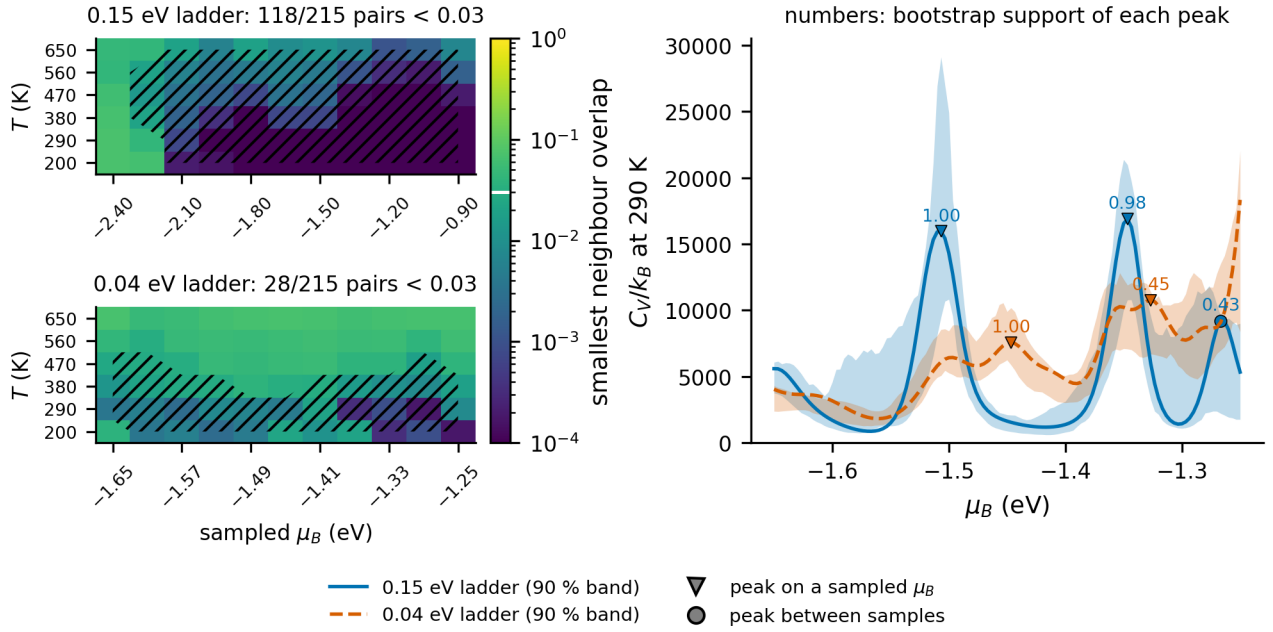


Figure 13. Reliability of the two LJ runs. Left: smallest MBAR overlap of each replica with a neighbour (hatched: below 0.03), on the coarse (top) and dense (bottom) ladder; the white line on the colourbar marks 0.03. Right: C_V at 290 K with 90 % block-bootstrap bands (60 resamples); triangles mark peaks on a sampled μ_B, circles peaks between samples, and the numbers are their bootstrap support.

Two further checks need no extra code:

- **Split the run**: compare the maps from the two halves. `samples.discard(0.5)` keeps the second half; for the first half, build a `SampleSet` from the first `n_samples // 2` entries of each replica.
- **Cells with $n_{\text{eff}} \approx 1$** have no meaningful average, whatever their error bar says.

5.5 Filling the gaps instead of starting over

The dense run of §4.1 was a second, separate run of 66 replicas. When the overlap check finds gaps, a cheaper route is to add states only where they are missing and analyse old and new samples together. MBAR does not need the states to form one rectangular grid.

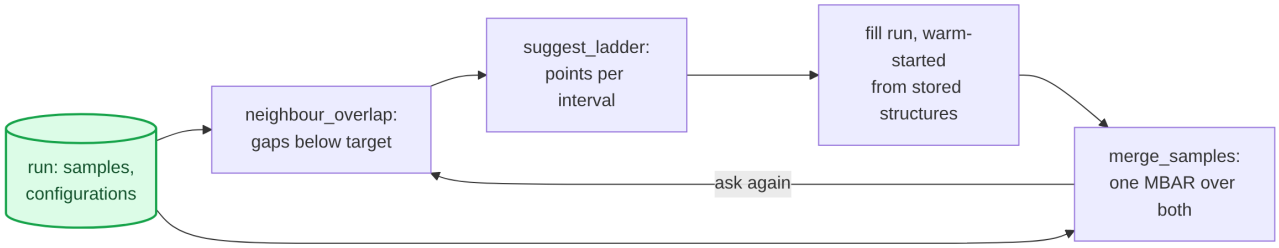


Diagram 6. Filling ladder gaps. The overlap of neighbouring replicas shows the gaps; the ladder advisor proposes new states there; a fill run starts from stored structures; its samples are merged with the original run for one MBAR analysis, and the advisor is asked again.

- **Advisor.** `suggest_ladder(analysis, target_overlap)` cuts every gap with overlap o into $\lceil \ln o / \ln o_{\text{target}} \rceil$ pieces. Each interval of the ladder gets one set of points, shared by all temperatures, so the fill runs are rectangular grids. `to_config()` writes them as `replica_exchange` snippets.
- **Merge.** `merge_samples(run, fill)` keys states `(run, it, imu)` and lists their (T, μ). It refuses runs that do not sample the same ensemble (exchanged species, Λ^3, substrate). In a config file, `phase_diagram.merge:`

`[fill_dir]` does the same, and also checks the structure and the potential.
- **Warm start.** `system_factory_from_configurations` starts every new replica from the last stored structure of its nearest sampled state.

For the coarse run of §4.1 the advisor proposes 17 new μ columns at a target overlap of 0.03 (102 replicas, about 1.5 times the original run). Merging the coarse run with the dense run of Figure 7 (132 states, solved in 17 s by first solving each run alone) cuts the neighbouring pairs below 0.03 from 118 of 215 to 36 of 231. Inside the dense window the merged ⟨N⟩ and C_V at 290 K lie within the dense run's 90 % bootstrap band at 81 % and 87 % of the points. Most of them fall at μ above −1.5 eV, where the coverage changes fastest. That is where the dense run of Figure 7 was needed, and the advisor's spacing there, about 0.04 eV, is the one chosen by hand for the dense run.

5.6 Another potential, without a new run

A REGC run is expensive; evaluating a potential on a stored configuration is not. If the configurations of a run with potential A are kept, the phase diagram of another potential B (a refitted EAM, a fine-tuned MLIP, single points from DFT) follows by reweighting, as long as B favours configurations that A visited.

The physics. MBAR gives every sample n drawn at the sampled states k the mixture density $\sum_k N_k e^{f_k - u_k^A(x_n)}$. The weight of the sample in any other ensemble is the ratio of that ensemble's Boltzmann factor to the mixture density [Shirts & Chodera]. For the grand ensemble of potential B at (T, μ):

$$w_n^B(T,\mu) \propto \frac{\exp[-\beta U_B(x_n) + \beta\mu N_n - N_n \ln \Lambda^3]}{\sum_k N_k \exp[f_k - u_k^A(x_n)]}.$$

Only $U_B(x_n)$ is new: the denominators are A's, solved once. Three consequences:

- **The effective sample size measures the overlap of A and B.** $n_{\text{eff}} = 1/\sum_n w_n^2$ under B, compared with A's own, says how many of A's samples still carry B's average. A ratio near 1 means B's distribution lies where A sampled; a ratio far below 1 means B favours configurations that A rarely visited, and only a run with B can say what happens there.
- **Shifts linear in N are free.** $U_B = U_A + cN$ is A at $\mu - c$, and the reweighting reproduces this to round-off (a unit test). What limits the overlap are changes in site preference and in the H–H interaction, not a change of the average binding energy.
- **Grand potentials.** The ratio of the two grand partition functions is an average over A, $e^{-\beta(\Omega_B - \Omega_A)} = \langle e^{-\beta(U_B - U_A)} \rangle_A$ at each (T, μ): free-energy perturbation [Zwanzig] in the grand ensemble (`grand_potential_shift`).

μ keeps its meaning only if both potentials put the gas reference (the isolated molecule: D_0, e_{atom}) at the same energy. If B binds H2 differently, its reservoir maps p to a different μ; pass B's reservoir to `with_potential` (`regc-reevaluate` stores it in the records file, and the runner uses it).

The records file. `records: {save: true}` (or `ReplicaExchangeGCMC(records_path=...)`) writes `records.h5`, with one row per recorded sample:

Dataset	Shape	Content
`samples/state`	(n, 2)	grid point (it, imu)
`samples/sample`	(n,)	sample index at that grid point
`samples/walker`	(n,)	the configuration's walker (followed through swaps)
`samples/N`, `samples/N_s`	(n,), (n, S)	mobile atoms (per species with several)
`samples/E/<name>`	(n,)	energy under potential `<name>`; NaN where not evaluated
`config/offset`, `config/length`	(n,)	where the sample's atoms are in `positions` (−1: not stored)

Dataset	Shape	Content
`config/positions`, `config/species`	(m, 3), (m,)	all stored atoms, frozen, substrate, mobile

The atom count varies from sample to sample, so positions are one long array addressed by offset and length. The file is flushed at every checkpoint and every 2048 records. After a resume, the samples recorded after the last checkpoint are written again, and readers keep the last record of each (state, sample), so the records always match `samples.npz`. In the Fe–H benchmark below (about 42 atoms per sample) the file takes 1.1 kB per sample in float64; `every: k` stores only every k-th configuration and `positions: float32` halves the size.

The workflow.

```
regc-run run.yaml                                          # records: {save: true}
regc-reevaluate run.yaml other_potential.yaml --name B --workers 8
# add  potentials: [B]  to phase_diagram, then
regc-run run.yaml --analyze-only    # <plot>_B.png, _B_difference.png, _B_ess.png
```

`other_potential.yaml` holds a `calculator` section (and any other section to replace). The new setup must build the same cell; if the lattice constant comes from the potential, fix it in `structure`. Re-evaluation writes after every chunk and skips what is done, so it can be interrupted and repeated, and it evaluates only the new samples after a run has been extended. `--every k` evaluates every k-th sample; the subset must not depend on the configurations, or the reweighting is biased. In Python:

```
from regc.records import reevaluate, attach_energies
reevaluate("out/records.h5", make_system_B, "B", n_workers=8)
samples = attach_energies(load_samples("out/samples.npz", discard=0.2),
 ↪  "out/records.h5")
A = MBARSurfaceAnalysis(samples, reservoir=res, pressure_unit="bar")
B = A.with_potential("B")                  # every map of B; bootstrap(B, ...) works too
```

Benchmark: H on Fe(100), EAM → EAM′. To test the method against a direct answer, the Fe–H EAM of [Wen] (A) was compared with EAM′ (B), the same potential plus a weak Fe–H Morse term, $D[(1-e^{-\alpha(r-r_0)})^2-1]$ with D = 0.01 eV, α = 1.5 Å^{-1}, r_0 = 2.0 Å, cut at 3 Å. EAM′ binds adsorbed H a few hundredths of an eV more strongly. Fe–Fe and H–H are unchanged, so the lattice constant (2.8549 Å), slab, H2 reservoir (D_0 = 4.4577 eV) and μ ladder are identical. Both potentials were sampled with the same setup:

Setting	Value
Slab	Fe(100) 3 × 3 × 4, 2 frozen + 2 displaceable layers, 9 surface atoms, H in z ∈ (0, 7.85] Å
Grid	400, 500, 625, 780 K × p_{H_2} = 10^{-6} … 100 bar (8 values) = 32 replicas
Sampling	REGC, x_0 = 0.9, MC diffusion, 15 000 steps, 1200 samples per replica kept
Cost	EAM 2 h 10 min, EAM′ direct 2 h 25 min (6 workers each); re-evaluating 48 000 EAM configurations with EAM′: 77 s

Before comparing, every 25th stored configuration of each run (1920 per run) was re-evaluated from scratch with its own potential. The energies recorded during sampling agree to 7.6 × 10^{-12} eV, so the direct EAM′ run sampled exactly the model that the re-evaluation computed.

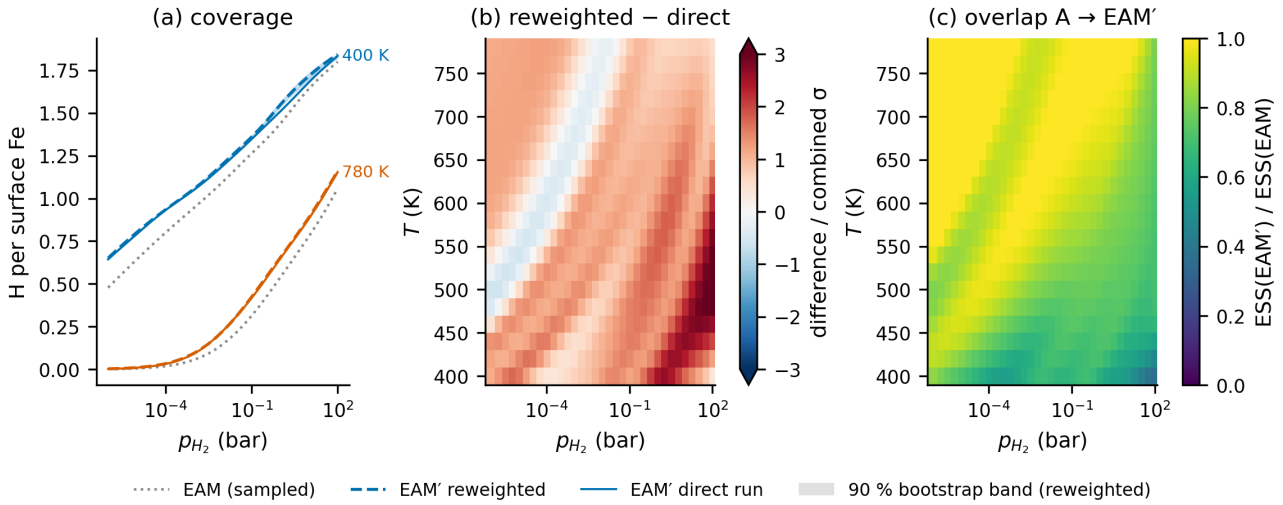


Figure 14. H on Fe(100): the EAM run reweighted to EAM′ against a direct EAM′ run. (a) Coverage at 400 and 780 K: EAM as sampled (dotted), reweighted to EAM′ (dashed, with its 90 % bootstrap band, 40 resamples) and the direct EAM′ run (solid). (b) Reweighted minus direct in units of the combined bootstrap standard error, z. (c) Effective sample size of the EAM samples under EAM′ divided by that under EAM.

Quantity (20 T × 33 p mesh, H per surface Fe)	Value
Change caused by EAM′ (direct runs)	up to 0.17, mean 0.09
Reweighted – direct	mean |d| 0.010, max 0.074
Bootstrap standard error, median	0.005 (reweighted), 0.006 (direct)
Cells with |z| < 2 / < 3	94 % / 99 %, max |z| 3.3
Mean z	+1.0
Share of the EAM ⟶ EAM′ change recovered	median 1.07 (5–95 %: 0.98–1.26), 527 cells with a change > 0.05
n_{eff} under EAM′ / under EAM	median 0.91, min 0.37; $n_{\text{eff}} \geq 759$ everywhere
$\Omega_B - \Omega_A$ on the sampled states	−0.001 eV (780 K, 10^{-6} bar) to −0.54 eV (400 K, 100 bar)

What the benchmark shows:

- **The change is reproduced.** EAM′ raises the coverage by up to 0.17 H per surface Fe. The reweighted map recovers that change to about 10 %, within 3.3 combined standard errors in every cell, from 77 s of energy evaluations instead of a 2.4 h run (Figure 14a, b).
- **Overlap is good everywhere** (Figure 14c). It is lowest at 400 K and high pressure. There the surface holds the most H, the per-atom change adds up (the grand potentials differ by 0.54 eV), and B's weights concentrate on fewer of A's samples. A stronger perturbation would first fail there.
- **A small systematic offset remains.** The reweighted coverage is higher than the direct one by about 0.01 H per surface Fe, about one combined standard error (Figure 14b is mostly red). Where it comes from:
  - It is not a code error: the energy check rules that out.
  - It is not poor overlap: with at least 759 effective samples, the bias of the reweighting ratio is far smaller.
  - It appears in both halves of the runs independently (+0.011 from samples 20–60 %, +0.009 from 60–100 %). Both runs creep upwards from the first to the second half, the direct EAM′ run more (+0.005 vs +0.003), so slow equilibration of the direct run explains part of it.
  - The rest is within what one pair of runs can show. The bootstrap ignores the correlation between replicas, and this ladder is coarse: 26 of 94 neighbouring pairs overlap less than 0.03, and `cross_left` swaps are accepted in 6.5 % of attempts (never for several pairs). Independent repeats with other seeds would measure the run-to-run scatter directly; this benchmark does not have them.

- **In practice**: read the effective-sample-size map first. Treat differences of about one bootstrap standard error between potentials as unresolved, and confirm the region that matters with a short direct run of the other potential.

Not yet implemented: one MBAR over states sampled with *different* potentials (runs of A and of B analysed together, each sample evaluated with both). The records file already holds what that needs.

6. Running from a config file

The command `regc-run config.yaml` runs the whole pipeline: structure → reservoir → REGC → MBAR → maps and structures. A minimal config with the new analysis options:

```
structure:
  build: {surface: fcc111, symbol: Pd, size: [4, 4, 4]}
  n_frozen_layers: 2
  headroom_angstrom: 6.0
gas: {species: O, insertion_z_range_above_surface: [-0.5, 3.5]}
calculator: {type: LAMMPS, pair_style: "grace/runtime",
             pair_coeff: "* * model.npz Pd O", species: [Pd, O], lammps_name: grace}
reservoir: {type: diatomic, d0: self, bond_length_angstrom: 1.2075,
            vibrational_wavenumber_cm: 1580.2}       # electronic_degeneracy 3 by default
 ↪  for O
replica_exchange:
  scheme: regc
  temperatures_K: [300, 400, 500, 650, 800]
  pressures: [1.0e-12, 1.0e-9, 1.0e-6, 1.0e-3, 1.0]
  pressure_unit: bar
  n_steps: 20000
  x0: 0.9
  diffusion: md
  diffusion_steps: 40
  md: {timestep_fs: 1.0, taut_fs: 20.0}
  record_every: 10
  adsorbed_cutoff: 2.4
  n_workers: auto
observables:
  - {type: coordination, name: cn_OPd, r_cut: 2.6}
  - {type: rdf, name: rdf_OO, r_max: 5.0, n_bins: 100, a: mobile, b: mobile}
configurations: {save: true, every: 2}
checkpoint: {chunk: 1000}
phase_diagram:
  T_min_K: 300
  T_max_K: 800
  n_T: 51
  p_min: 1.0e-12
  p_max: 1.0
  n_p: 61
  observable: N_ads
  most_probable: true
  heat_capacity: true
  min_ess: 20
  rdf_states: [[400, 1.0e-6], [700, 1.0e-6]]
  export_configurations: {mode: phases, min_cells: 4}
  bootstrap: {n: 100, seed: 0}                    # intervals, label agreement, C_V peak
 ↪  support
output: {directory: output}
```

- `--check` builds and prints the setup (structure, reservoir, μ ladder, energies) without sampling.

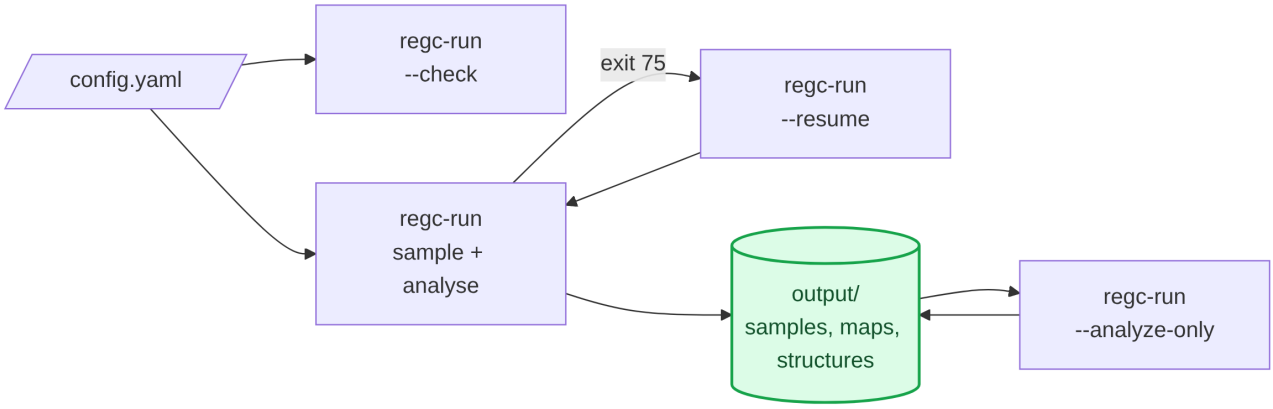


Diagram 7. regc-run command modes. The check mode builds and prints the setup without sampling, a normal run samples and analyses, a resumed run continues from the checkpoint, and analyze-only recomputes maps and exports from saved samples.

- A plain run samples and analyses; with a `checkpoint` section it stops cleanly with exit code 75 at a time limit.
- `--resume` continues from the checkpoint.
- `--analyze-only` recomputes the maps and exports from the saved samples, for example with a new mesh, `min_ess` or export mode.

Output	Contents
`samples.npz`	U, N, N_ads, observables per replica and sample; ladder, μ grid, Λ^3, observable definitions
`run_metadata.json`	the full setup (resume and `--analyze-only` read it)
`phase_diagram.png`, `_most_probable.png`, `_heat_capacity.png`, `_cn_<name>.png`, `_rdf_<name>.png`	the maps
`phase_diagram_data.npz`	all maps as arrays, with `effective_samples`, the C_V ridge points, the reliability arrays (`n_independent`, `overlap_min`, `round_trips_T`) and, with `bootstrap`, the intervals and agreement
`phase_diagram_uncertainty.png`	with `bootstrap`: half-width of the coverage interval
`mixing.json`	swap acceptance per pattern and per pair, and round trips per walker (§5.4)
`configurations/configs_T*_mu*.extxyz`	sample-tagged structures
`configurations_phases.extxyz`, `..._tiled.extxyz`	the structures of the phase diagram

Every key is listed in the reference manual, §14.

---

7. Long runs on a cluster

`run_with_checkpoints` and `regc-run` save `samples.npz` and a pickled checkpoint after every chunk. Each file is written atomically: a temporary file is synced to disk, then renamed. A resumed run continues bit-identically. After every chunk the driver also checks each replica's running energy against a full recomputation, and resyncs it, with a warning, if it has drifted.

Resuming is refused when anything that defines the samples has changed:

- the T ladder, the μ grid or Λ^3;
- `adsorbed_cutoff` or the observables;
- any key of `run_metadata.json`, which is compared in both directions.

The calculators, however, are taken from the current setup. A moved potential file still works, and a changed potential setting takes effect, with a warning if the stored energies change. Samples recorded before such a change belong to the old potential: discard them (or restart), or MBAR will mix two energy surfaces.

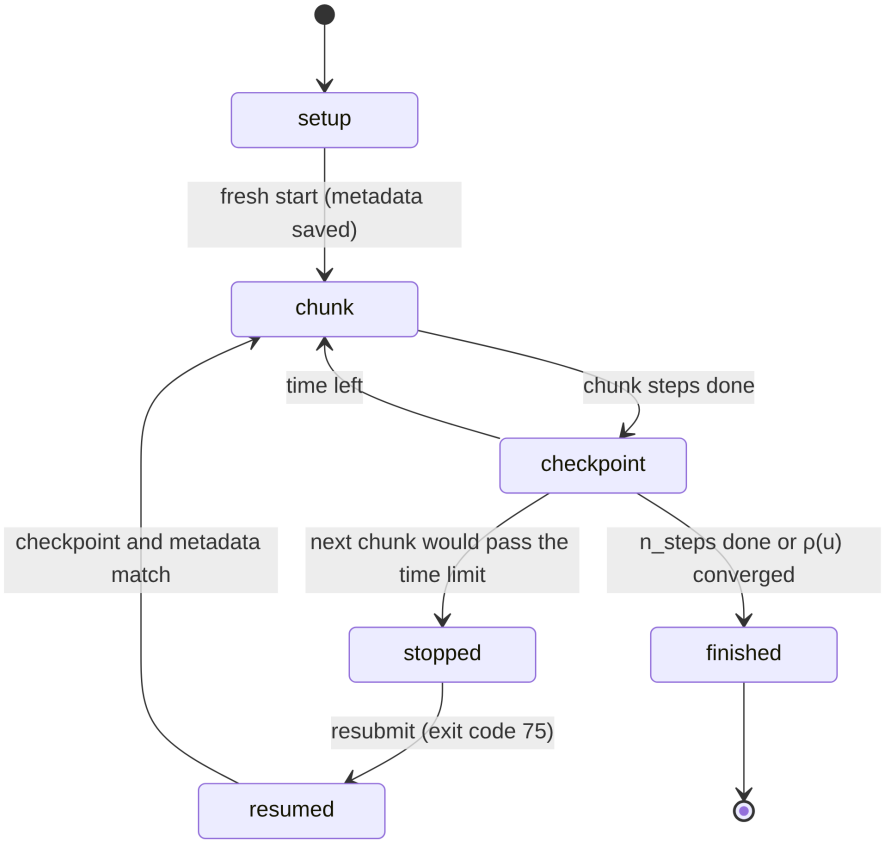


Diagram 8. Life cycle of a checkpointed run. A run is set up, then alternates sampling chunks and checkpoints; a time limit ends the job with exit code 75 for resubmission, a resumed run reloads the checkpoint, and the run ends when the steps are done or the convergence test passes.

`8050:localhost:8050 node`). The page loads Plotly from a CDN in the browser, or from a local file with `--plotly-js`.

---

8. Good practice and pitfalls

Check	Why	How
Units	T enters as k_BT in eV; Λ^3 must be the exchanged atom's	`kelvin_to_eV`, `reservoir.lambda_cubed_of_T()`
D_0 and e_atom from the potential	an external D_0 shifts every μ by the potential's error	`DiatomicReservoir.from_calculator(..., d0="self")`
Swap acceptance 20–40 %	lower: replicas do not mix; higher: wasted replicas	`sim.swap_acceptance()`, printed every chunk
Equilibration discarded	early samples remember the start	`discard=0.2`; check that 0.4 gives the same maps
Effective samples	maps beyond the ladder are extrapolations	`effective_sample_size`; `min_ess: 20`
Independent samples $n/g \geq 20$ per replica	correlated samples overstate what a run knows	`replica_inefficiencies`; printed by `regc-run`

Check	Why	How
Neighbour overlap ≥ 0.03	between poorly overlapping replicas, variances such as C_V are biased low	`neighbour_overlap`; add replicas in the gaps
C_V peaks off the ladder	peaks on the sampled μ or p are an overlap artefact (§4.1)	`heat_capacity_ridges(..., ladder=...)`; fill the gaps with `suggest_ladder` and analyse with `merge_samples`
Disjoint-block convergence	a nested comparison always "converges"	`convergence_every` with blocks long enough for the noise floor
Cutoffs below half the cell	otherwise periodic images are missed	errors are raised; enlarge the cell
Insertion zone covers the adsorption sites	atoms outside it cannot be deleted	`insertion_z_range`, `insertion_energy_scan`
p_insert = p_delete	detailed balance	enforced
Quantum nuclei	H is classical here: no zero-point energy on the surface	compare with D, or correct afterwards
Finite cell	every N step gives a C_V bump; long-period phases need a larger cell	compare two cell sizes

When something goes wrong:

- **Nothing adsorbs:** check the sign and scale of μ against e_atom, and the insertion zone.
- **Everything adsorbs:** check D_0 (an experimental D_0 with an approximate potential) and Λ^3.
- **Swap acceptance near 0:** neighbouring T or μ are too far apart for their energy distributions to overlap; add replicas.
- **Maps change with `discard`:** the run is not equilibrated; run longer.

---

9. References

- **[ZSG19]** Y. Zhou, M. Scheffler, L. M. Ghiringhelli, *Determining surface phase diagrams including anharmonic effects*, Phys. Rev. B **100**, 174106 (2019). doi:10.1103/PhysRevB.100.174106
- **[ZSG22]** Y. Zhou, C. Zhu, M. Scheffler, L. M. Ghiringhelli, *Ab initio approach for thermodynamic surface phases with full consideration of anharmonic effects: the example of hydrogen at Si(100)*, Phys. Rev. Lett. **128**, 246101 (2022). doi:10.1103/PhysRevLett.128.246101
- **[Reuter–Scheffler]** K. Reuter, M. Scheffler, *Composition, structure, and stability of RuO_2(110) as a function of oxygen pressure*, Phys. Rev. B **65**, 035406 (2001). doi:10.1103/PhysRevB.65.035406
- **[Shirts & Chodera]** M. R. Shirts, J. D. Chodera, *Statistically optimal analysis of samples from multiple equilibrium states*, J. Chem. Phys. **129**, 124105 (2008). doi:10.1063/1.2978177
- **[Bussi]** G. Bussi, D. Donadio, M. Parrinello, *Canonical sampling through velocity rescaling*, J. Chem. Phys. **126**, 014101 (2007). doi:10.1063/1.2408420
- **[JANAF]** M. W. Chase Jr., *NIST-JANAF Thermochemical Tables*, 4th ed., J. Phys. Chem. Ref. Data Monograph 9 (1998); tables H-050 (H_2) and O-029 (O_2), https://janaf.nist.gov.
- **[Klimovich]** P. V. Klimovich, M. R. Shirts, D. L. Mobley, *Guidelines for the analysis of free energy calculations*, J. Comput.-Aided Mol. Des. **29**, 397–411 (2015). doi:10.1007/s10822-015-9840-9
- **[Wen]** M. Wen, *A new interatomic potential describing Fe-H and H-H interactions in bcc iron*, Comput. Mater. Sci. **197**, 110640 (2021). doi:10.1016/j.commatsci.2021.110640
- **[Zwanzig]** R. W. Zwanzig, *High-temperature equation of state by a perturbation method. I. Nonpolar gases*, J. Chem. Phys. **22**, 1420–1426 (1954). doi:10.1063/1.1740409
- **[Künsch]** H. R. Künsch, *The jackknife and the bootstrap for general stationary observations*, Ann. Statist. **17**, 1217–1241 (1989). doi:10.1214/aos/1176347265
- **[SAS]** T. Kassis, V. Agarwal, Y. He, D. Patel, A. M. Brueckner, *Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents*, arXiv:2609.00065 (2026). doi:10.48550/arXiv.2609.00065

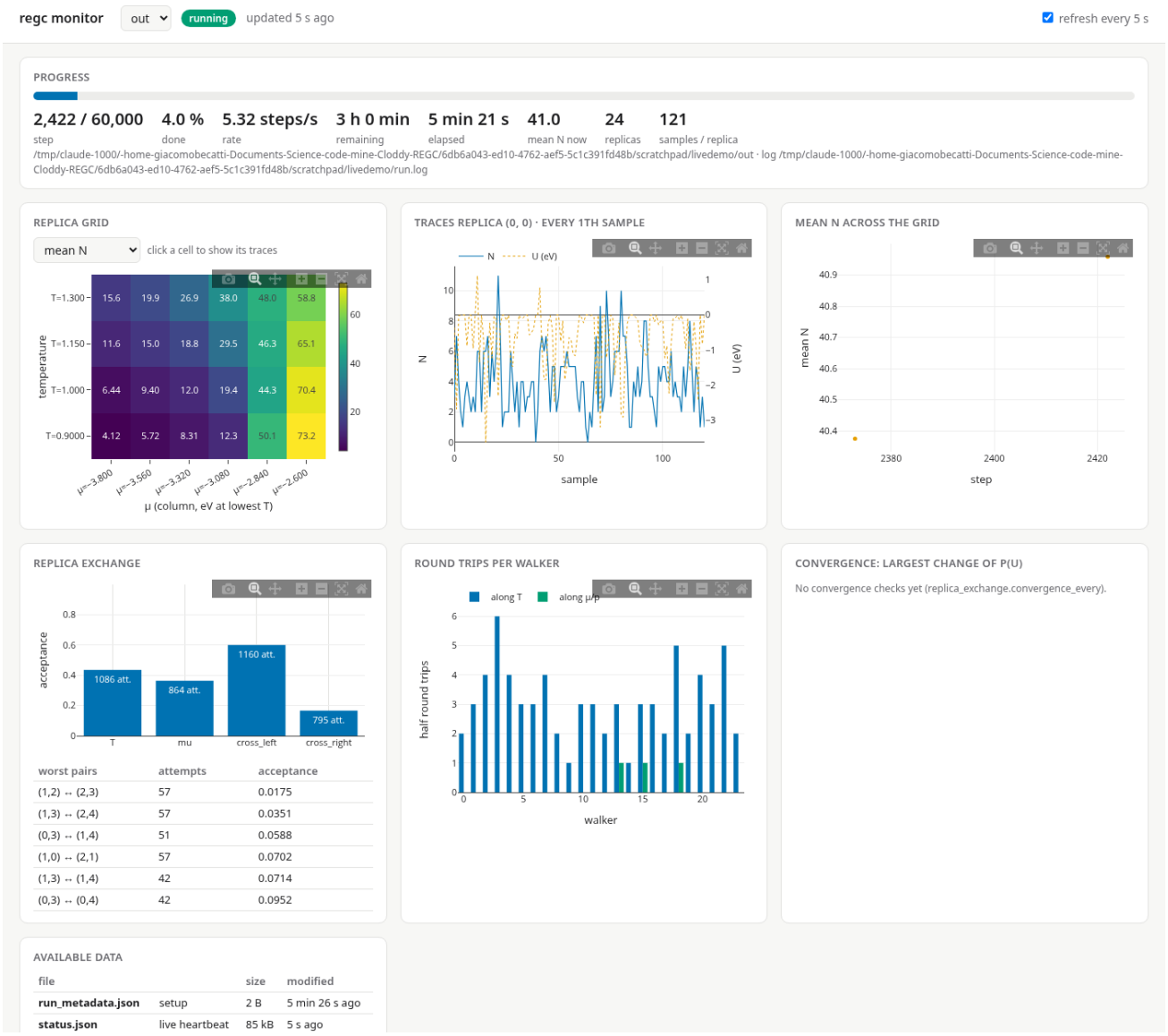


Figure 15. The live monitor on a running LJ test run (24 replicas, 2 422 of 60 000 steps): progress with rate and remaining time, mean N on the replica grid with one replica's N and U traces, and swap acceptance with the worst neighbouring pairs and round trips.

Appendix A. Figure provenance

Every figure is produced by `docs/manual/make_figures.py` with the package in this repository; diagrams are Mermaid sources inside this file (exported to `figures/diagrams/*.mmd` and rendered by `build_manual.py`).

Figure	Function	Data	Settings
1	`fig_reservoirs`	`diatomic_gas_mu_eV`; JANAF values from `tests/test_reservoir.py`	H_2, O_2 spectroscopic constants of §1.3
2	`fig_swap_patterns`	`swap_pairs` of the engine	4×5 grid
3	`fig_ideal_gas`	a sweep-scheme run, seed 7	$V = 27$, $T = 1.0, 1.5, 6$ μ, 1500 samples per replica

Figure	Function	Data	Settings
4	`fig_convergence`	synthetic Gaussian series, seed 1	4000 samples per block, 10 bins
5, 6, 9–12	`fig_lj_surface`, `_draw_phases`	coarse `examples/14` run (§4.1), first 20 % of each replica discarded	320 samples per replica; mesh 46 T × 61 μ; $n_{\mathrm{eff}} \geq 20$
7, 8	`fig_cv_ladder`, `fig_lj_surface`	coarse and dense `examples/14` runs (`--mu-min -1.65 --mu-max -1.25 --mu-step 0.04`)	as above; dense map 46 T × 41 μ
13	`fig_reliability`	both runs	`neighbour_overlap`; `bootstrap` with 60 resamples, seeds 0 and 1, 90 % intervals
14	`fig_reweighting`	`data_feh_reweighting.npz`: the Fe–H EAM and EAM′ runs of §5.6 (seed 0 each), first 20 % discarded	mesh 20 T × 33 p; `bootstrap` with 40 resamples, seed 1, for each map
15	screenshot	`regc-monitor` on an LJ test run (4 T × 6 μ, seed 0), headless Chrome	—

The LJ runs are short illustrations: no error bars are shown on their maps, and the effective sample size (Figure 9) is the reliability measure used to mask cells.

The LJ figures are regenerated with `python docs/manual/make_figures.py --run-dir <coarse> --dense-run-dir <dense>`.

This manual was drafted and revised with an AI assistant using the Scientific Agent Skills library [SAS] (scientific writing, visualization, diagramming and critical-review procedures). Every number in it was taken from the code, its tests, or the runs listed above, and two independent review passes checked the claims against the code and the cited papers. The maintainers are responsible for its content.